Counting Processes, Regression, and Bayesian Methods for Epidemiology

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Prompt: Conditioning, Regression and Models for Count Data

Abstract

We will first explore how to model the spread of disease through a population, focusing on the Galton-Watson branching process for the early stages of an outbreak and the SIR model for later stages. Turning this section, the idea of treating infectious periods as a random variable from a Poisson process is also explored, and the modelling implications discussed. Once the branching process has been modelled, with discuss several techniques for estimating the parameter R_0 from data collected from various pandemics and epidemics. These techniques include objective Bayesian parameter estimation, linear regression (best linear predictor), and Bayesian linear regression. Then, how R_e changes overtime due to mitigation efforts is explored using Bayesian A/B testing. Finally, we will look at the efficacy of Gaussian naive Bayes' classifiers and Bayesian logistic regression for prediction of disease.

Contents

1 Modelling the Spread of Disease.

1.1 Early Stages: The Galton-Watson Process.

The spread of disease, especially the early stages of epidemics, is easily modeled using branching processes—namely, the Galton-Watson process. A branching process is a stochastic process, and the Galton-Watson has two distinct features: a) all individuals give birth according to the same probability law independently of each other, and b) the number of offspring produced by an individual is independent of the number of individuals in that generation [4].

With respect to epidemiology, the number of individuals in generation n, X_n, is equivalent to the number of **new cases** during a given infectious period, and offspring refers to the number of new people, $Y^(j)_k$ each individual in a generation infects, where j refers to the generation and k to the individual of that generation. So, clearly

$$X_{n+1} := Y_1^{(n)} + Y_2^{(n)} + ... + Y_{X_n}^{(n)}$$

[23].

It is not unreasonable to model the number of offspring using a Poisson distribution [22]:

$$Y_K^{(j)} \in Po(\lambda)$$

And in fact, this simplifies some of the math. When discussing the spread of disease, one of the relevant variables of interest is called R_0. This value, R_0, is the **basic reproductive number**, and is a metric used to describe the transmissibility of a disease [28]. Referring back to Ahlberg's paper

$$R_0 = EY$$

which, in the case of the Poisson distribution, is simply the Poisson parameter, λ. In other words,

$$R_0 = EY = \lambda$$

Intuitively, this makes sense. The offspring, $Y_k^{(j)}$, really just represents the number of infectious contacts of k^{th} individual in generation j. So, if we think about the number of infectious contacts as being Poisson distributed, then we already know that λ is the mean number of infectious contacts, by definition of the Poisson parameter.

1.2 Late Stages: The SIR Model.

Interestingly, it is well documented that probabilistic models, like the Galton-Watson branching process, are really only necessary in the **beginning** stages of an outbreak. As time progresses, the simpler SIR model– which is a deterministic ODE model– is used instead as it fits data just as well and is less computationally dense [23][20]. In the SIR model, "S" stands for **susceptible**, "I" for **infected**, and "R" for **recovered**. From Smith and Moore of the Mathematical Association of America [18], we therefore know we can model an outbreak of disease over time as the following:

$$s(t) = \text{proportion of population that is susceptible at time t.}$$

$$i(t) = \text{proportion of population that is infected at time t.}$$

$$r(t) = \text{proportion of population that is recovered at time t.}$$

And the differential equations related to the SIR model are as follows:

$$\frac{ds}{dt} = -b \cdot s(t)i(t)$$

$$\frac{dr}{dt} = k \cdot i(t)$$

$$\frac{di}{dt} = b \cdot s(t)i(t) - k \cdot i(t)$$

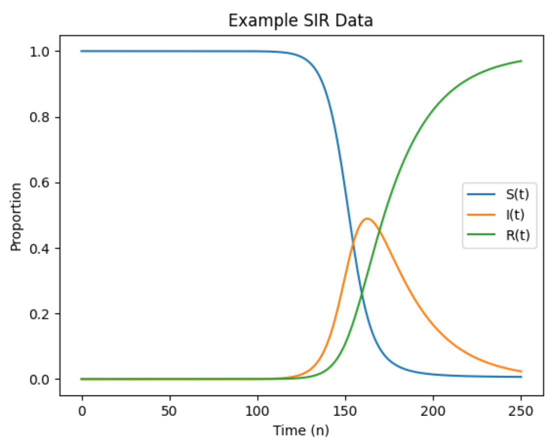
Where b is the proportion of infected-susceptible interactions per day, that result in infection (i.e., that are sufficient to spread the disease) and k is the fraction of the infected group that will recover during a given generation. For the purposes of example, k will equal one. Again, intuitively, these equations make sense. Of course the number of individuals leaving susceptible status is proportional to -b, the number of susceptible people and the number of infected people, at time t. And then, of course, the number of infected increases by the same, and decreases by the number of infected at time t, when $k = 1$. Then, the number of recovered is just the number of people leaving infected status at time t.

A natural question is how the parameters b and k relate to R_e or R_0. We can define R_e as follows

$$R_e = \frac{b \cdot s(t)}{k}$$

So, R_e is **decreasing** overtime, and when once $R_e < 1$, the infected population will decrease to 0[24]. We also see this in Allan Gut's *An Intermediate Course in Probability*[4], which states η, the probability of (eventual) extinction is 1 when $EY_k^{(j)} < 1$. Something interesting to note about this model is that the parameters b, k, and $i(0)$ are frequently determined by a trial and error method [24]. Looking ahead to the

Figure 1

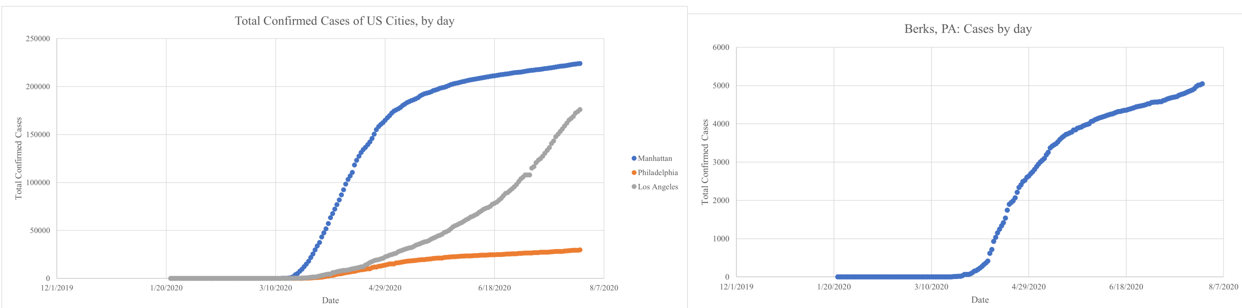


section where we work out R_e for various counties in the US, we can see the orange line in Figure 1, $I(t)$, shares a similar shape to what we see in the "New Cases by day" charts, as we would expect. Figure 1 also provides some insight into the relationship between $S(t), I(t)$, and $R(t)$.

1.3 COVID-19 Data By County.

The data from the CDC and CSSE at Johns Hopkins [41] gives us the total number of confirmed cases of COVID-19 and deaths by COVID-19 as a function of date [41]. The data starts on January 22, 2020 and ends on July 27, 2020. During this timespan, the CDC tracked these numbers by county for every county in the United States. It is also important to note that thought the CDC begins tracking cases on January 22, 2020, the first case in each county varies. This becomes important when accounting for generation, n.

For the purposes of this analysis we'll choose several counties from throughout the country to work with individually. Those counties will be Berks County, Pennsylvania, Manhattan County New York, Los Angeles County California, and Philadelphia County, Pennsylvania. Figure 2 shows the cumulative cases for each of these specified counties:



(a) Total Confirmed Cases Major US Cities, by day

(b) Total Confirmed Cases Berks, PA, by day

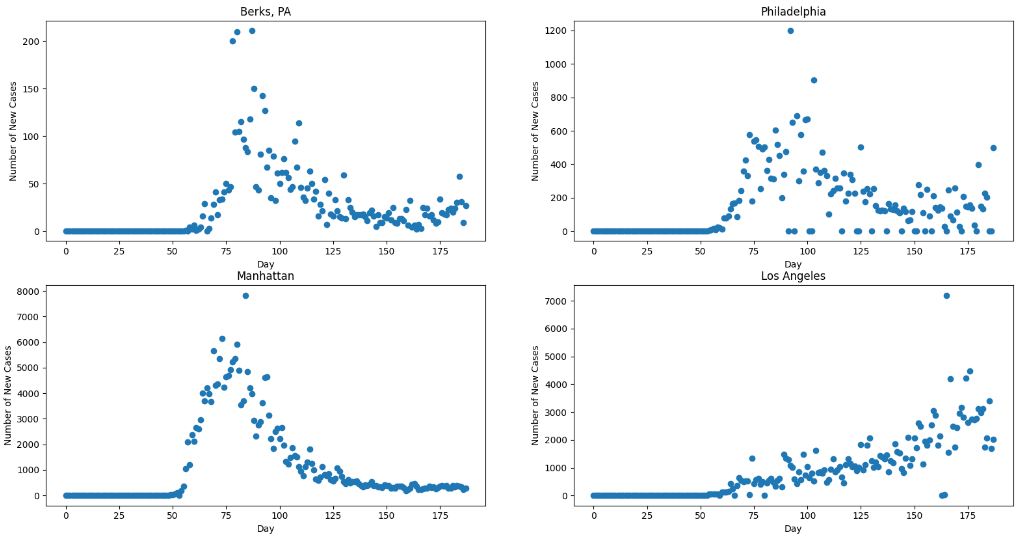
Figure 2: Total Confirmed Cases, by county

Really, what the above graphs are showing is T_i– that is, the total number of cases each day, cumulatively. What we are really interested in is $X(i)$– or the number of new cases each generation. Because the data is taken each day, we simply find X_i to be

$$X(i) = T_i - T_{i-1}$$

The CDC data was transformed this way and $X(i)$ for the various US counties of interest are plotted below: What's interesting about each graph, is that even though three of them have very similar shapes the scales are drastically different. Not to mention one county, that being Los Angeles, actually has a very different shape than the other three counties in terms of $X(n)$ versus N. This was a little surprising to see, since we would have expected similar R_0 values for each county considering we're dealing with the same disease in each. But there are some rational reasons why this could have occurred. For one, access to testing resources could have artificially deflated the number of detected cases in certain areas. Of the four counties that we've chosen, Berks County is by far the smallest and least densely populated. So in densely populated counties like Los Angeles or Manhattan, you would expect to see more interactions per day therefore driving up the effective R_0, or R_e. This could explain why those two counties have many more cases in pure numbers than places like Berks county.

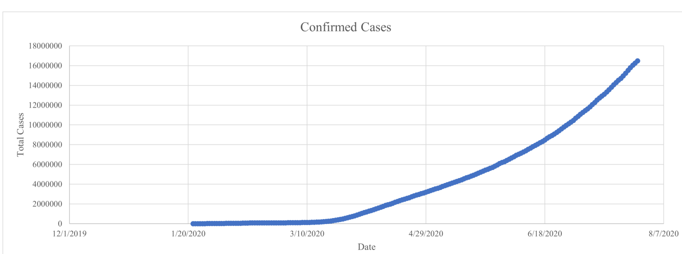
Figure 3: New Cases Major US Counties, by day



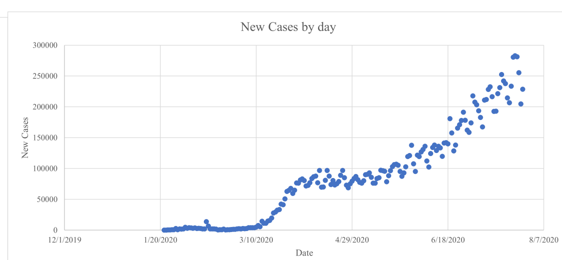
2 Data Sets.

2.1 COVID-19 Data Worldwide.

The CDC and the CSSE at Johns Hopkins [41] also tracked the number of cumulative cases and cumulative deaths by day world wide similarly to what was done by county in the United States [41]. From this data, using a similar method as before, we will again determine R_0– but this time, for the entire world. Again, the data is visualized below:



(a) Total Confirmed Cases Worldwide, by day



(b) New Confirmed Cases Worldwide, by day

Figure 4: Total Cases & New Cases of COVID-19 World Wide.

2.2 Ebola Data Set.

Similar to our previous work examining the COVID-19 pandemic, we have our T_i data (i.e., the total rolling reported number of cases) for the 2014-2016 West Africa Ebola Outbreak [45]. Looking at the below graphs, we can see a few peculiarities:

- Though the data comes from the CDC, unlike COVID-19 in the US, the number of newly infected Ebola patients was not recorded daily, so there are some several day gaps in the data.
- At times, the total case count decreases by a few patients day-to-day. It is unclear how this is possible, and is not discussed on the CDC's site. Most likely, some previously suspected cases are ruled out as being Ebola.
- Given the date and location of this outbreak, if anything the number of reported cases is well under the true number of cases.
- Unlike the COVID-19 data that was granular to the scale of individual counties, the Ebola data is totalled across all three countries: Guinea, Sierra Leone, and Liberia.

Figure 5: Total Confirmed Ebola Cases in West Africa (2014-16), using the raw CDC data set.

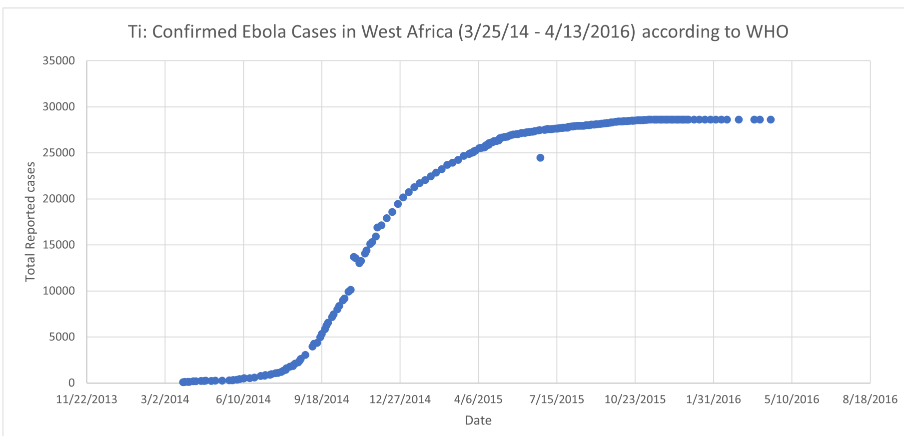
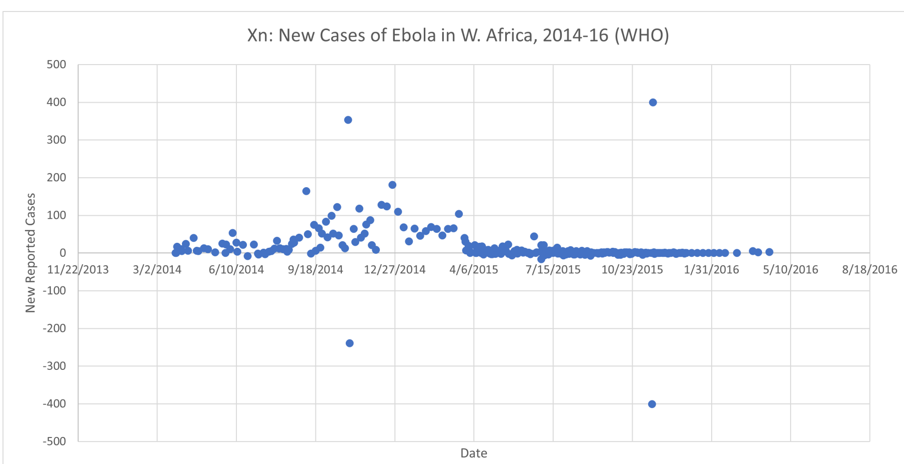


Figure 6: Newly Confirmed Ebola Cases in West Africa (2014-16), using the raw CDC data set.



3 Parameter Estimation Using Baye's Theorem.

3.1 What is R_0?

If you are at all familiar with epidemiology or the study of disease in general, then you've probably heard of the value R_0. Simply stated R_0 is the average number of people an infected person will infect during their infectious period. The value varies by disease, and it depends on many factors, but in general the higher the R_0 value the more likely the disease is to spread through a population.

In the context of this paper, we can think of R_0 as the expected number of children in the Galton-Watson process we will use to describe the spread of an epidemic. It is widely accepted that the number of infectious interactions a carrier will have is Poisson distributed. So, if we let $Y_k^{(j)}$ be the infected cases ("offspring") caused by individual k in generation j, then

$$Y_k^{(j)} \in Po(\lambda)$$

This relationship allows us to estimate $R_0 = \lambda$. Below are some of the currently (as of December 2023) accepted R_0 values for various strains of SARS-CoV-2 [27].

COVID-19 Strain	R_0
Alpha	1.22
Beta	1.19
Gamma	1.21
Delta	1.38
Omnicron	1.90

At the beginning of the pandemic, the literature has initial calculations for R_0 much higher and were much more varied. One Chinese study[25] compared 12 studies published in the first few months of 2020 and found R_0's ranging from 1.5 to 6.68. Especially in the early stages, under reporting due to lack of awareness could account for artificially low R_0s, then over reporting due to hysteria could account for artificially high R_0s. The great variation in early reports can be explained somewhat by the behaviors and mitigation method certain populations take. R_0 is the average number of infectious **interactions**, so societies that reduce their interactions via lockdown or other means will skew their local R_0 while societies that make no changes will see high R_0s. For example, an Italian model estimated R_0 from 2.76 to 3.25 initially but after mitigation measures saw a decrease in R_0 [28].

3.1.1 R_0 vs. R_e.

R_0 is the basic reproductive number and is the reproductive ratio that most people are familiar with, but there is also the **effective** reproductive number, R_e, that changes as the immunity of the population changes. Where you might consider R_0 to be the "true" reproductive ratio, R_e can be thought of as describing how human behavior, herd immunity, and other outside factors influence the spread of disease. For example, R_0 is affected purely by the infectiousness of the organism and the rate of recovery and death during an outbreak. While, R_e is affected by herd immunity, either through natural immunity of significant proportions of the population contracting the disease or through immunization efforts. Obviously, such efforts will affect the rate at which the disease spreads, but we don't describe this using R_0, we described this using R_e [28]. In essence, R_0 remains, regardless of human intervention.

Within the confines of this paper, we will be attempting to estimate R_0 values of diseases in various counties and locations. We will also be seeing how certain efforts, namely lockdown efforts, affect the spread of disease. In these cases, we will be estimating R_e and the effect that mitigation efforts have on R_e.

3.2 λ as a Random Parameter

In this section we will explore what happens when we consider the infectious period of a disease as a random variable.
Looking at the expression for infected offspring gleaned from the branching process discussed previously, recall:

$$Y_k^{(j)} \in Po(\lambda)$$

where $\lambda = R_0 = EY_k^{(j)}$ is the expected number of new infectious caused from the k^{th} individual in generation j. While using the average number of infections caused by an infectious individual is useful and convenient, in reality some people's immune systems will respond better than others. In the case of COVID-19, some people are completely asymptomatic, and unwittingly spread the infection to more people, while some people's immune systems will handle whatever disease more readily, decreasing their infectious period.

Therefore, our branching process becomes: the infectious periods, T, are independent, identically distributed random variables; the reproduction rate means that infectious individuals infect others at a constant rate of $\lambda(= R_0)$ throughout their lifetime [4]. Therefore, our Poisson parameter becomes:

$$\lambda T$$

and the number of offspring of each infected individual becomes

$$Y_k^{(j)}|T = \tau \in Po(\lambda\tau) \text{ where, } T \in Exp(1/\mu)$$

3.2.1 Infections as a Poisson Process.

Rather than simply describing $Y_k^{(j)} \in Po(\lambda T)$, we can also think of this situation as a Poisson process [4].

Let $\{Y_k^{(j)}(t), t \geq 0\}$ be the number offspring the k^{th} individual in generation j has at time t. The rate of infection for this process is λ, and the times between each newly infected offspring are $T \in Exp(1/\mu)$. This representation is equivalent to the branching process we previously described, and the number of offspring of each infected individual again becomes

$$Y_k^{(j)}|T = \tau \in Po(\lambda\tau) \text{ where, } T \in Exp(1/\mu)$$

and thus, the following derived distributions for $Y_k^{(j)}$ are applicable for either interpretation.

3.2.2 Exponentially Distributed Infectious Periods.

Using the aforementioned distribution of $y_k^{(j)}$, by the Law of Total Probability, where i is the number of infectees of the k^{th} individual of generation j:

$$P(Y_k^{(j)} = i) = \int_0^\infty P(Y_k^{(j)} = i|T = t) \cdot f_\tau(t) \cdot dt$$

$$= \int_0^\infty e^{-\lambda t} \cdot \frac{(\lambda t)^i}{i!} \mu e^{-\mu t} \cdot dt = \int_0^\infty \frac{\mu(\lambda t)^i}{i!} \cdot \exp(-[\lambda + \mu]t) \cdot dx$$

$$= \mu \int_0^\infty \frac{(\lambda t)^i}{\Gamma(i+1)} \cdot \exp(-[\lambda + \mu]t) \cdot dt$$

$$= \frac{\mu\lambda^i}{(\lambda+\mu)^{i+1}} \int_0^\infty \frac{(\lambda+\mu)^{i+1} t^i}{\Gamma(i+1)} \cdot \exp(-[\lambda + \mu]t) \cdot dt$$

We recognize the integral as the PDF for a $\Gamma(t + 1, 1/[\lambda + \mu])$, being integrated over its support. Therefore, by the Law of Total Probability this integral reduces to 1, leaving us with

$$P(Y_k^{(j)} = i) = \frac{\mu\lambda^i}{(\lambda+\mu)^{i+1}} = \frac{\mu}{(\lambda+\mu)} \cdot \left(\frac{\lambda}{(\lambda+\mu)}\right)^i$$

Which we recognize mean that $Y_k^{(j)}$ is a geometric random variable, specifically it is $Ge\left(\frac{\mu}{\lambda+\mu}\right)$-distributed.

In fact, we can take this one step further and look at the probability of non-extinction, $1-\eta$ [4]. By Theorem 3.7.3(c) in Allan Gut's *An Intermediate Course in Probability*, we know that if $m \leq 1$, them $\eta = 1$ and if $m > 1$, then $\eta < 1$. We also know that $m = EY_k^{(j)}$ [4], and because we have already established $Y_k^{(j)} \in Ge\left(\frac{\mu}{\lambda+\mu}\right)$, then

$$m = EY_k^{(j)} = \frac{q}{p} = \frac{\frac{\lambda}{\lambda+\mu}}{\frac{\mu}{\lambda+\mu}} = \frac{\lambda}{\mu}$$

So, we can see that we have two cases:
If $m > 1$, then $m = \frac{\lambda}{\mu} > 1$, that is, if $\mu > \lambda$, then $\eta = 1$, and therefore the probability of non-extinction is

$$1 - \eta = 0$$

Or, if $m \leq 1$, then $m = \frac{\lambda}{\mu} \leq 1$, that is, if $\mu \leq \lambda$, then we know by Theorem 3.7.3 (b) that η is the smallest non-negative root of

$$x = g_{Y_k^{(j)}}(x)$$

where $g_{Y_k^{(j)}}(x)$ is the probability generating function of $Y_k^{(j)}$.
Because $Y_k^{(j)} \in Ge\left(\frac{\mu}{\lambda+\mu}\right)$

$$g_{Y_k^{(j)}}(x) = x = \frac{p}{1-qt}$$

and therefore to solve this equation, recalling that $\lambda, \mu > 0$:

$$x(1-qx) = p$$

$$x\left(1 - \frac{\lambda}{\lambda+\mu}x\right) = \frac{\mu}{\lambda+\mu}$$

$$(\lambda+\mu)x\left(1 - \frac{\lambda}{\lambda+\mu}x\right) = \mu$$

$$\left((\lambda+\mu)x - \lambda x^2\right) = \mu$$

$$\lambda x + \mu x - \lambda x^2 = \mu$$

$$\lambda x - \lambda x^2 = \mu - \mu x$$

$$\lambda x(1-x) = \mu(1-x)$$

$$\lambda x = \mu$$

$$\eta = x = \frac{\mu}{\lambda}$$

and therefore in this case, the probability of non-extinction

$$1 - \eta = 1 - \frac{\mu}{\lambda}$$

This can be interpreted as probability that the outbreak of a disease is self limiting, or in other words, that an epidemic dies out on it's own. Note that a $1-\eta = 1$ means certainty that the outbreak is self-limiting, and the closer $1-\eta$ is to 1 the more likely it is to die out on it's own. Interestingly, because of the distribution of the infectious periods is $\tau \in Exp(1/\mu)$, then $E\tau = \mu$ means that as the average infectious period **decreases**,

$$1 - \eta \longrightarrow 1$$

assuming λ is constant and disease specific. This means that if we can manipulate the infectious period, we could ensure the extinction of an outbreak. As we will discuss (and see in our analysis) later, if lock-downs our other preventive measures are imposed to lessen the period of time an individual is effectively infectious, this idea rings true.

3.2.3 Uniformly Distributed Infectious Periods.

We have no reason to believe the infectious periods, T, are exponentially distributed. So, as an alternative example, let

$$T \in U(0, \frac{b}{\lambda})$$

where $\frac{b}{\lambda}$ represents the maximum possible (or observed) infectious period.
Again, by the Law of Total Probability,

$$P(Y_k^{(j)} = i) = \int_0^\infty P(Y_k^{(j)} = k | T = t) \cdot f_\tau(t) \cdot dt$$

$$= \int_0^\infty e^{-\lambda t} \frac{(\lambda t)^i}{i!} \cdot \frac{1}{b} dt$$

using the equality $\Gamma(i+1) = i!$, we see

$$= \int_0^\infty e^{-\lambda t} \frac{(\lambda t)^i}{\Gamma(i+1)} \cdot \frac{1}{b} dt$$

$$= \frac{1}{b} \cdot \frac{\lambda^i}{\Gamma(i+1)} \int_0^\infty t^i e^{-\lambda t} dt$$

Recalling that $\int_0^\infty t^i e^{-1 \cdot t} dt = \frac{\Gamma(i+1)}{\lambda^{i+1}}$, this expression evaluates down to

$$= \frac{1}{b} \cdot \frac{\lambda^i}{\Gamma(i+1)} \cdot \frac{\Gamma(i+1)}{\lambda^{i+1}}$$

$$P(Y_k^{(j)} = i) = \frac{1}{b \cdot \lambda}$$

or, rather

$$Y_k^{(j)} \in U(0, b \cdot \lambda)$$

.
Intuitively, this $Y_k^{(j)} \in U(0, b \cdot \lambda)$ makes little sense, as it implies that $Y_k^{(j)}$ is a continuous random variable, which it clearly is not. Rather, this exercise was to show that depending on the distribution of people's infectious period for various diseases, the distribution of $Y_k^{(j)}$ varies.
Moving forward, we will continue to assume that the early stages of a pandemic are well modelled by the typical Galton-Watson process and that

$$Y_k^{(j)} \in Po(\lambda(= R_0)).$$

3.2.4 Probability of Extinction, η, & Conclusion.

In subsequent analyses throughout this project, we will treat the infectious periods as a constant number of days across all individuals. So, again we will be left simply with

$$Y_k^{(j)} \in Po(\lambda(= R_0))$$

We have already established

$$m = EY_k^{(j)} = \lambda(= R_0)$$

So, by Theorem 7.3 (c) in Allan Gut's book [4], we know if

$$\lambda(= R_0) < 1$$

then the probability of (eventual) extinction, η, is

$$\eta = 1$$

In other words, if $R_0 < 1$, the epidemic or outbreak will be self-limiting ad will not be able to proliferate. and

3.3 Estimating R_0 using Baye's Theorem.

Bearing all this in mind, we will try to estimate R_0 using Baye's Theorem during the early stages of the pandemic. Using CDC data which was collected daily from January 22, 2020 to July 27, 2020 that contains the number of confirmed cases and confirmed deaths by county in the US.

First, lets' talk about estimating Poisson Parameters. Let n be the generation, then we can think of each generation as being the **infectious period** of each infected person. From the most recent guidance form the CDC, we know that each generation, n, is 5 days [26][30].

Let $X(n)$ be the number of newly infected people in a generation. Then

$$X(n) = Y_1^{n-1} + Y_2^{n-1} + ... + Y_{X(n-1)}^{n-1}$$

where $Y_k^{(j)}$ is infectees ("offspring") of the k^{th} individual from the $n-1$ generation, and $X(n)$ is the total number of infected people in generation n. It is very typical to estimate the number of offspring as i.i.d Poisson random variables:

$$Y_k^{(j)} \in Po(\lambda)$$

Therefore, R_0 is the expected value of $Y_k^{(j)}$, which from Gut's textbook we know to be

$$R_0 = E[Y_k^{(j)}] = \lambda$$

Dr. Jarad Niemi of Iowa State University has many lectures on this topic. Below, is a synthesized version of his lectures on estimating Poisson parameters, applied specifically to the case of evaluating R_0 [5].

Let y be the observed confirmed cases of SARS-CoV-2 in the CDC dataset and recall $R_0 = \lambda$. Baye's Theorem tells us

$$P(\lambda|y) = \frac{P(y|\lambda) \cdot P(\lambda)}{P(y)}$$

or even more simply

$$P(\lambda|y) = C \cdot P(y|\lambda) \cdot P(\lambda)$$

where $C = \frac{1}{P(y)}$ is a proportionality constant. This leaves us with:

$$P(\lambda|y) \propto P(y|\lambda) \cdot P(\lambda)$$

Because we have assumed all $Y_k^{(j)}$ to be independent, we can represent $P(y|\lambda)$ as the **likelihood**, which is product of the individual observations:

$$P(y|\lambda) = \prod_{i=1}^{n} \frac{\lambda^{y_i} \cdot e^{-\lambda}}{y_i!}$$

Where y_i represents the observed value infectees for a given observed infected individual. Again using proportionality,

$$P(y|\lambda) \propto \lambda^{\sum^{n} y_i} \cdot e^{-n\lambda}$$

So, we can see that the likelihood, $P(y|\lambda)$ takes the form corresponding to $y|\lambda \in \Gamma(\sum^{n} y_i, \frac{1}{n})$. So, we can therefore choose $\lambda \in \Gamma(\alpha, \frac{1}{\beta})$ as our **conjugate** prior, where a conjugate prior is simply one that has the same type of distribution as the posterior. In fact, we can find the form of the posterior simply by

$$P(\lambda|y) = C' P(y|\lambda) \cdot P(\lambda)$$

$$P(\lambda|y) \propto P(y|\lambda) \cdot P(\lambda) = \lambda^{\alpha + \sum^{n} y_i - 1} \cdot e^{-(\beta+n)\lambda}$$

So, clearly our posterior is given by

$$\lambda|y \in \Gamma(p = \alpha + \sum^{n} y_i, a = \frac{1}{\beta + n})$$

Then, from Gut's textbook, because $\hat{R_0}$ is Gamma-distributed, we know

$$\hat{R_0} = E\lambda|y = p \cdot a = \frac{\alpha + \sum^{n} y_i}{\beta + n}$$

In the next section, we will apply this to the CDC'c by county COVID-19 data, for selected US counties and world wide.

3.4 Assumptions & Methodology.

3.4.1 COVID-19.

So, we have our $X(i)$ data for each county, as a function of 1 day steps. This would imply, if $X_i = X_n$, that

$$n = \text{generation} = 1 \text{ day}$$

With the power of hindsight, we know that this is not the case. The current CDC guidance is that after symptoms occur, most people are infectious for the next 5 days [26]. So, we should instead have:

$$n = \text{generation} = 5 \text{ days}$$

This is an easy enough fix, and we can use the T_i data for each county to calculate the appropriate X_n, given the generation length. This is simply

$$X(n) = T_i - T_{i-5}$$

where $n = 1$ for each county is with respect to the first detected case in that county. The data looks similar in shape, but note the peaks are higher: Recall form before that

$$EX(k) = \lambda^k$$

for a branching process, and because we are assuming the number of infectious interaction to be $Y_k^{(j)} \in Po(\lambda)$, we therefore have

$$EY = \lambda = R_0$$

and, thus,

$$EX(k) = R_0^k$$

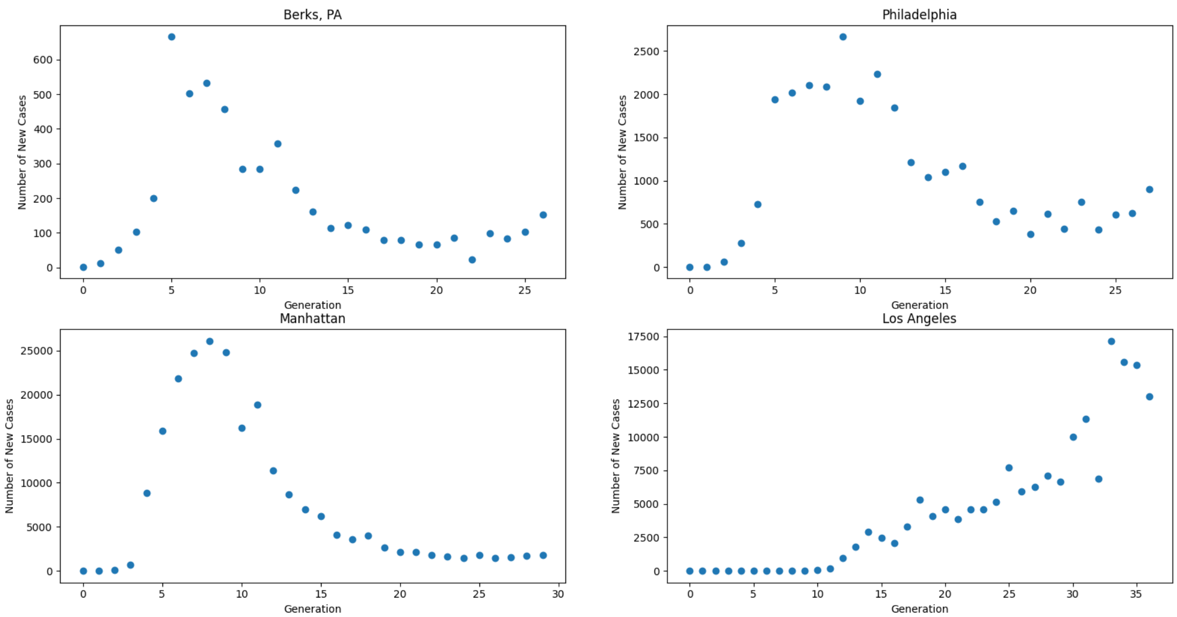
We can use the data we have graphed above to estimate R_0 for each generation, k. We will denote the observed $X(k)$ as x_k, and because for each county we only have 1 data point at each generation, $x_n = \overline{X(k)} \approx EX(n)$ in this case. The interesting thing is, that even though we only have 1 $X(n)$ for each $n \geq 1$ for each county, because we have several generations and we are assuming $Y_k^{(j)} \in Po(\lambda = R_0)$, we end up with a point of estimate of $\hat{R_0}$ per generation, per county. Also, let the estimators for λ– which we will call y_k– are simply derived by

$$y_k = (x_k)^{\frac{1}{k}}$$

for each observed generation k. In other words, if we let x_k be our observed newly infected at each generation of $n = 5$ days, then y_k will be the observed R_0 at that generation.

Those y_k can be seen in Figure 8.

Figure 7: $X(n)$: New Cases Major US Counties, n = 5 days



Let our prior distribution for λ be

$$\lambda \in \Gamma(1,1) \stackrel{d}{=} Exp(1)$$

and we can use our expression for the expectation of the posterior that we derived earlier

$$\hat{R_0} = E\lambda|y = p \cdot a = \frac{\alpha + \sum^{n} y_i}{\beta + n}$$

to estimate R_0 for each county interest, assuming $n = 5$, $\alpha = 1$, and $\beta = 1$. A summary of these results (including varied generation duration) is in Table 1.

Generation days	**Berks**	**Philadelphia**	**Manhattan**	**Los Angeles**
1	1.072	0.948	1.12	0.82
5	1.776	1.946	2.335	1.152
10	2.996	3.796	5.481	1.687

Table 1: $\hat{R}_0$ values for various generation lengths (i.e., infectious periods) in various US counties.

Figure 8: y_k: $\hat{\lambda}$ estimations by generation, by county, for n = 5 days

Generation (n)	Berks	Philadelphia	Manhattan	Los Angeles
1	3.606	1.732	3.162	0.000
2	3.708	3.979	4.380	0.000
3	3.186	4.072	5.178	0.000
4	2.888	3.738	6.156	0.000
5	2.955	3.530	5.016	0.000
6	2.431	2.966	4.168	0.000
7	2.191	2.602	3.541	0.000
8	1.975	2.338	3.096	1.318
9	1.759	2.201	2.751	1.302
10	1.672	1.989	2.414	1.466
11	1.632	1.902	2.272	1.554
12	1.516	1.783	2.052	1.693
13	1.438	1.661	1.911	1.707
14	1.371	1.589	1.805	1.703
15	1.350	1.549	1.727	1.630
16	1.319	1.515	1.631	1.567
17	1.274	1.445	1.577	1.569
18	1.259	1.391	1.547	1.571
19	1.233	1.382	1.483	1.516
20	1.222	1.328	1.442	1.494
21	1.224	1.339	1.417	1.456
22	1.144	1.304	1.385	1.443
23	1.211	1.318	1.361	1.421
24	1.193	1.275	1.339	1.407
25	1.195	1.279	1.334	1.411

3.4.2 Ebola.

For our assumptions, we have know that Ebola is not infectious until the onset of symptoms and it is also accepted that once symptoms emerge, and untreated Ebola victim will dies in about 10 days [40]. Therefore, the infectious period (or generation length) is

$$n = generation = 10 \text{ days}$$

and, recalling our notation that i represents days, while n represents generations,

$$X(n) = T_i - T_{i-10}$$

Something else that is relevant in this case is that we are likely missing data regarding the earlier generations of this outbreak. As aforementioned, the outbreak was believed to have originated from bats in Guinea in December of 2013 [38], but the data from the CDC [44][45] begins on March 25, 2014– with an initial count of 86 Ebola cases. We have already established $n = 10$ days, this means that from December 31, 2013 to March 25, 2014 at least 5 generations have passed, as far as our Galton-Watson model is concerned. So, we will adjust accordingly.

Again, recall that we are assuming the number of offspring Ebola cases caused by an infected individual to be Poisson distributed, $Y_k^{(j)} \in Po(\lambda)$. This means,

$$EX(k) = R_0^k$$

where, to account for the missing generations, $k \geq 5$.

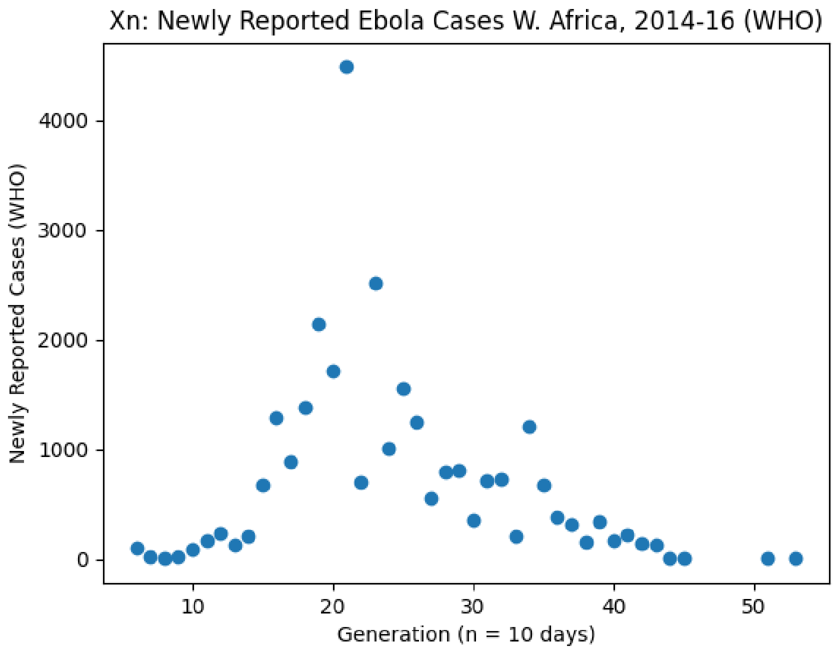
Now, using the same prior and estimation of $\hat{R_0}$ we did in our exploration of the COVID-19 Data:

$$\lambda \in \Gamma(1,1) \stackrel{d}{=} Exp(1)$$

$$\hat{R_0} = E\lambda|y = p \cdot a = \frac{\alpha + \sum^{n} y_i}{\beta + n}$$

we can also make an estimate for R_0 of the Ebola outbreak in West Africa, varying some of the assumptions we have already made:

Figure 9: $X(n)$, after making reasonable assumptions about the CDC data.



Generation (days)	Initial generation	$\hat{R}_0$
1	**1**	0.528
5	**1**	2.180
10	**1**	3.811
1	**3**	0.292
5	**3**	0.943
10	**3**	1.293
1	**5**	0.287
5	**5**	0.897
10	**5**	1.183

Table 2: $\hat{R}_0$ values for various generation lengths (i.e., infectious periods) in West Africa.

3.4.3 Discussion of Results.

Methodology Discussion. This method is only feasible if the infectious period for each individual is known– as we can see, the estimated $\hat{R}_0$ depends quite a bit on this interval. Too low, and we may underestimate R_0, and too high and we may overestimate R_0. Though, another way to think of the infectious period = 1 day case, is to think of it as the average number of people an infectious person infects in a day– though, they maybe infectious for longer.

Ebola Discussion. As previously mentioned in our discussion of R_0 versus R_e, the way the public attempts to mitigate the spread of disease has major implications on the resultant R_e value. An interesting note, in the case of Ebola, is that unlike COVID-19, Ebola is spread only through contact with infected fluids [38]. This means one can't contract the disease through coughing or other aerosolized methods of transmission. This means that all of the cases of Ebola in western Africa from 2014 to 2016 came directly from physical contact with an infected person's viscera.

The initial patient was believed to be a toddler from a village in Guinea that was infected by bats, during December of 2013 [38]. From there, it was able to spread into neighboring countries Sierra Leone and Liberia because of the increased mobility of people in West Africa compared to previous Ebola outbreaks, and the fact that their were no early detection systems for the disease and health care workers in the area were untrained on diagnosing the ancient illness [39].

It is reasonable to wonder how the disease spread so rapidly, given most people would avoid someone with Ebola-like symptoms. This is where the unique customs of western Africa shaped the effective reproduction, R_e, of Ebola in 2014. In the region, there is a strong "adherence to ancestral funeral and burial rites singled out as fuelling large explosions of new cases" [39]. Medical anthropologists in the region had even previously noted that the areas funeral practices were exceptionally high risk.For example, in Liberia and Sierra Leone "some mourners bathe in or anoint others with rinse water from the washing of corpses" [39]. Bathing in the viscera of a descendant of Ebola explains partially how Ebola was able spread so rapidly in the region. Other factors include severe shortage of medical workers, unfamiliarity with the disease, and the reliance of "traditional" healers over modern medical practitioners all being prevalent in the area.

As for the actual results, we can see why making reasonable assumptions is crucial if this method is to be used to determine R_0 for Ebola. Clearly, R_0 cannot be less than 1, otherwise an outbreak would not occur, as not enough people would be infected– this is why determining an appropriate generation length in days is needed. As we can see, if the generation duration is too low, in this case if we assume it to be 1 or 5 days, we get $R_0 < 1$, which is clearly false. Determining how far into outbreak we are is also paramount: we are assuming $X(1) = 1$, so if we underestimate how many generations of the disease have already propagated, then we will **overestimate** our R_0. For example, in the case of this Ebola data, the first data point we have is 86 confirmed cases. The assumption $X(1) = 1$, means if we assume $X(2) = 86$ as well, we will get

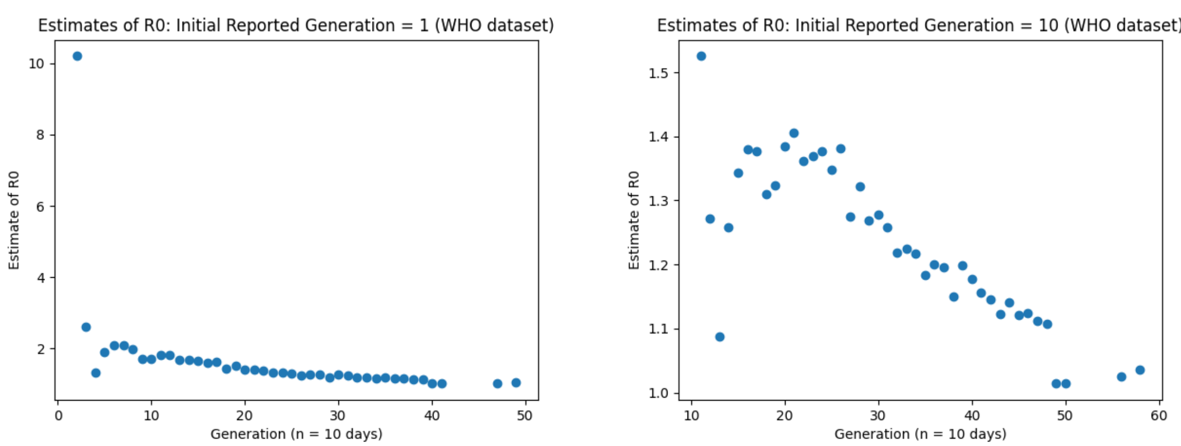
$$\hat{R_0} = 9.27$$

as our first estimate. But, correcting for the missing early generations, assume $X(6) = 86$ will instead yield

$$EX(6) = 86 = \hat{R_0}^6$$

$$\hat{R_0} = 2.1$$

Interestingly, the estimates of $\hat{R_0}$ do seem to level out over time, as seen in Figure 10.



(a) (b)

Figure 10: Estimates of $\hat{R_0}$ for Ebola.

So, given our assumptions, assuming the first generation we have data for is actually $n = 5$ and that the generation length is 10 days, $\hat{R_0} = 1.183$ is our estimate for R_0 of Ebola. The accepted value is anywhere between 1.4-2.0, depending on the outbreak and the methods used for estimation [37][38].

4 Parameter Estimation Using the Best Linear Predictor.

From Allan Gut's *An Intermediate Course in Probability*[4], we know $h(\mathbf{x})$ is called a regression line if

$$h(\mathbf{x}) = h(x_1, ..., x_n) = E(\mathbf{Y}|X_1 = x_1, ..., X_n = x_n) = E(Y|\mathbf{X} = \mathbf{x})$$

This means in order to find the best linear predictor

$$L(X) = \beta + \alpha X$$

that is, the one that minimizes $E(Y - (\beta + \alpha X))$, where α, β are constants, we know [4]

$$\beta = \mu_y - \frac{\sigma_{xy}}{\sigma_x^2}\mu_x = \mu_y - \rho\frac{\sigma_y}{\sigma_x}\mu_x$$

$$\alpha = \frac{\sigma_{xy}}{\sigma_{xy}^2} = \rho\frac{\sigma_y}{\sigma_x}$$

and thus

$$\hat{Y} = \mu_y + \rho\frac{\sigma_y}{\sigma_x}(X - \mu_x)$$

where

$$\mu_y = \text{mean of observed responses.}$$

$$\mu_x = \text{mean of observed independent variable(s).}$$

$$\sigma_y^2 = \text{variance of observed responses.}$$

$$\sigma_x^2 = \text{variance of observed independent variable(s).}$$

$$\rho = \frac{Cov(X, Y)}{\sqrt{VarX \cdot VarY}} = \text{correlation coefficient.}$$

4.1 Estimating R_0 using Regression.

4.1.1 COVID-19

Perhaps a better way to estimate our parameter of interest, R_0, it's just simply take advantage of the fact that the expected value of $Y_k^{(j)}$ is in fact λ. We also know from Theorem 7.2 in Gut's text [4] that the expected value of each generation in a Galton-Watson branching process is simply

$$EX(n) = (EY)^n$$

where n is the generation.

We can use the same data as in our previous method, but representing it differently, we can instead use **linear regression** to determine and estimate of R_0. So, we can rewrite:

$$EX(n) = (EY)^n$$

$$EX(n) = \lambda^n = R_0^n$$

In our data, let x_n be the observed value of X_n in each generation n, $n \geq 1$, and let $\hat{R_0}$ be the observed value of R_0. We therefore have

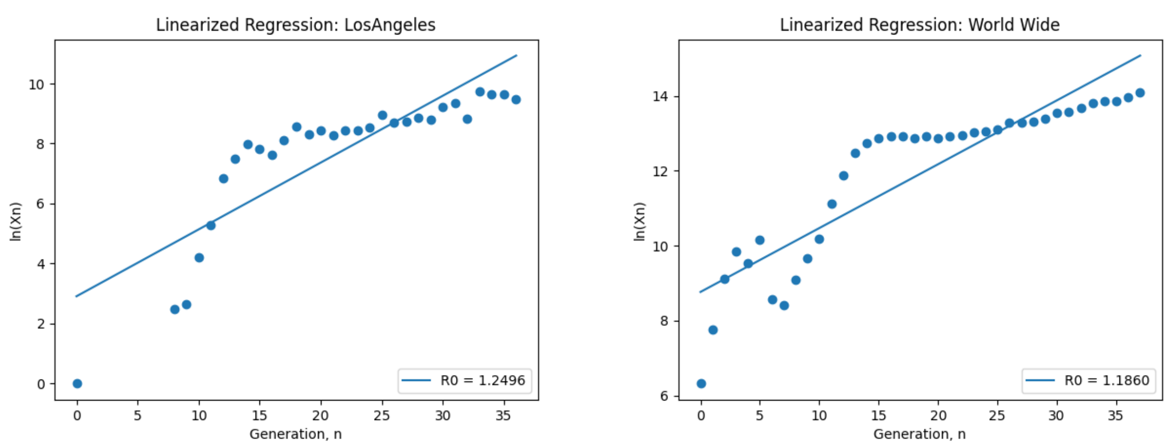
$$x_n = \hat{R_0}^n$$

We can easily linearize this by taking the natural logarithm of each side of our observed data for each county:

$$\ln(x_n) = n \cdot \ln(\hat{R_0})$$

Notice that this is in the $R = \alpha + \beta \cdot x$ format. Therefore, the slope, m, of our transformed data can be used to find $\hat{R_0}$ as follows

$$\hat{R_0} = e^m$$

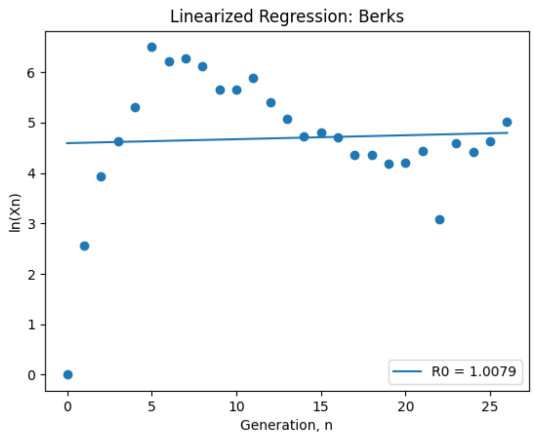


(a) Estimating R_0 in Los Angeles county, CA, from 1/22/20-7/27/20

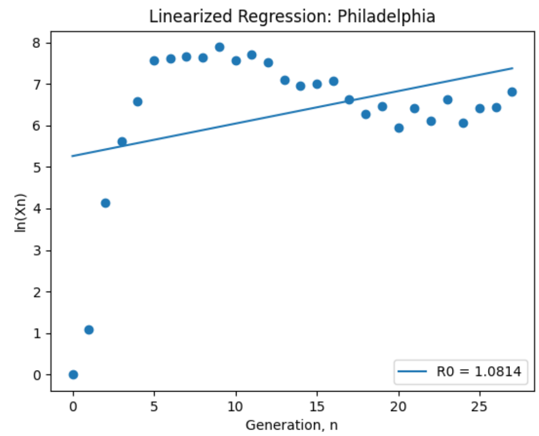
(b) Estimating R_0 from all confirmed cases from around the world.

Figure 11: Best Linear Predictor Regression Analysis on COVID-19 Data.

In some cases, what it actually looks like is **two** linear functions, with different slopes. Around the third generation in both the cases of Philadelphia county and Berks county, something happens where the steep incline originally seen in cases suddenly drops off. This begs the questions: **did lock-downs decrease** R_0? Based on the graphs it seems so– in fact, the slopes seem to be **negative** after generation 4, and a negative slope on these graphs implies an $R_e < 1$. We will explore this more in the next section, again using Bayesian A/B Testing to see if λ before and after lock-downs is actually different in subsequent sections.



(a) Estimating R_0 in Berks county, PA, from 1/22/20-7/27/20



(b) Estimating R_0 in Philadelphia county, PA, from 1/22/20-7/27/20

Figure 12: Best Linear Predictor Regression Analysis on COVID-19 Data.

4.1.2 Ebola.

In the case of Ebola, the distinction between R_0 and R_e becomes even more relevant and visible. As we can see in the figure above, we can see a distinct drop in the slope somewhere around the 20th generation (about 200 days, so late November or early December of 2014 in real time). Recall, that the relationship between R_0 (or rather R_e) and the slope, m, is

$$m = \ln(\hat{R}_0) \iff \hat{R}_0 = e^m$$

So, the sudden decrease in slope, m, reflects a decrease in $\hat{R}_0$. In fact,

$$m < 0 \iff \hat{R}_0 < 1 \iff \eta = 1$$

In fact, if we split the data at the 20th generation and examine the resultant $\hat{R}_0$ values, we see that before the 20th generation there is an **increasing** relationship in between $\ln X(n)$ and n, with a calculated value of

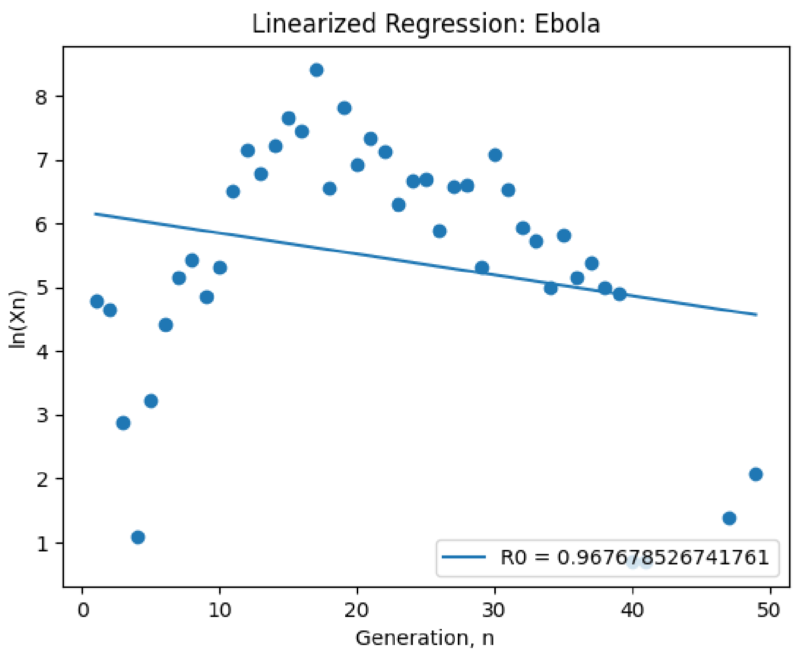
$$\hat{R}_0 = 1.20$$

Versus, after the 25th generation a fairly linear **decreasing**relationship in between $\ln(X_n)$ and n, with a calculated value of

$$\hat{R}_0 = 0.739$$

From Ahlberg's paper [23] and Gut's book [4], we know that $\hat{R}_0 < 1 \iff \eta = 1$ means the outbreak will die out. Looking at the data in Figure 13, this again begs the question of whether or not human intervention **decreased** R_e after generation 20, in the case of the 2014-2016 Ebola outbreak in West Africa.

Figure 13: $\hat{R}_0$ estimation for for all generations (3/25/2014-4/13/2016)



4.2 Bayesian Linear Regression.

Instead of simply employing the Best Linear Predictor (BLP), we can use Bayesian Linear Regression. As the name suggests, this is a method that instead of delivering point estimates of the regression parameters α (slope), and β (intercept), will deliver posterior distributions for these parameters. We will explore and implement two different methods of Bayesian Linear Regression.

4.2.1 Objective BLR.

From Dr. Jarad Niemi of Iowa State University's [6] lecture on objective Bayesian linear regression, we can, in the multivariate case, define our regression problem as

$$\mathbf{Y} \in N(\mathbf{X}\beta, \sigma^2 I)$$

where $\mathbf{Y}, \mathbf{X}$ are our response variable and independent variables, respectively, of the forms

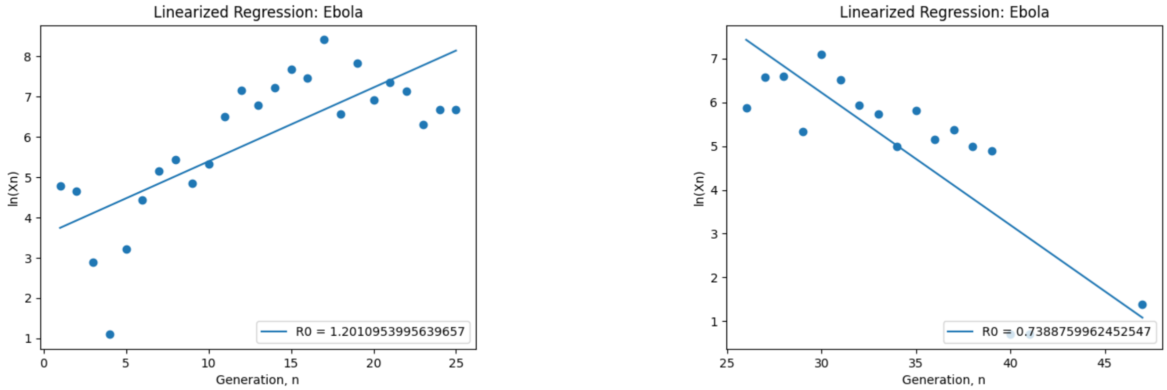
$$\mathbf{Y} = (y_1, y_2, ..., y_n)$$

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} 1 & x_{12} & x_{13} & \dots & x_{1k} \\ 1 & x_{22} & x_{23} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots & \\ 1 & x_{n2} & x_{n3} & \dots & x_{nk} \end{bmatrix}$$

where n is the number of observations, and k the number of independent variables.
We are also assuming that our $\mathbf{X}_i$'s are independent random variables, there is a constant variance, σ^2, and that $\mathbf{X}_{i1} = 1$ are our intercepts.

We will choose a non-informative constant prior

$$p(\beta, \sigma^2) \propto \frac{1}{\sigma^2}$$



(a) $\hat{R_0}$ estimation for first 25 generations (3/25/2014-11/30/2014)

(b) $\hat{R_0}$ estimation for 25th generation to end of outbreak (11/30/2014-4/13/2016)

Figure 14: Best Linear Predictor Regression Analysis on Ebola Data.

Then, using Baye's Theorem

$$p(\beta, \sigma^2|\mathbf{Y}) = \frac{p(\beta|\sigma^2, y) \cdot p(\sigma^2, y)}{p(\beta, \sigma^2)}$$

and because $p(\beta, \sigma^2)$ is proportional to a constant

$$p(\beta, \sigma^2|\mathbf{Y}) \propto p(\beta|\sigma^2, y) \cdot p(\sigma^2, y)$$

This leaves us with the following posteriors for our parameters β and σ^2:

$$\beta|\sigma^2 \in N(\hat{\beta}, \sigma^2 V_\beta)$$

$$\sigma^2|\mathbf{Y} \in Inv - Gamma(\frac{n-k}{2}, \frac{[s^2 n - k]}{2})$$

where, with the addition of QR-factorization of $\mathbf{X} = \mathbf{QR}$, where $\mathbf{Q}$ is an orthonormal matrix and $\mathbf{R}$ and upper right triangular,

$$\hat{\beta} = \mathbf{R}^{-1}[\mathbf{R}']^{-1} = \text{least squares estimator.}$$

$$V_\beta = \mathbf{R}^{-1}\mathbf{Q}'\mathbf{Y}$$

$$s^2 = \frac{1}{n-k}(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = \text{estimator of } \sigma^2.$$

4.2.2 Markov Chain Monte Carlo (MCMC) BLR.

The goal of this method is very similar to that of objective BLR: to determine and sample from the posterior distributions of β, α, and σ. But, the objective method relies on priors and posteriors that are well defined. The Markov Chain Monte Carlo (MCMC) method instead uses and algorithm to traverse and sample from uncommon posterior distributions.

Intuition. In the context of Bayesian Inference, the Metropolis-Hastings algorithm can be used to draw random samples from a posterior distribution. The algorithm works by simulation a Markov Chain– in essence, MCMC is given some initial value, x_1, then a new value, call it x_2 is proposed. The ratio between the two values is evaluated, and if the ratio is found to be acceptable it is kept, otherwise it is disregarded. Eventually, iterating through this algorithm, the samples will being to approximate the posterior distribution.

This is especially useful in Bayesian Inference, where the true posterior distribution may be difficult or impossible to analytically derive [7] [2]. As we draw more an more samples from the posterior distribution, it will begin to converge to the true posterior PDF– and this makes it simple for us to plat the distributions, and determine key statistics about our parameters of interest, such as mean and variance, to better understand what we actually know about the parameter.

Metropolis-Hastings. In the context of our exploration of R_0 values for COVID-19, the Metropolis-Hastings algorithm is the following [8]:

1. Begin with some initial initial parameter, call it λ^c
2. Evaluate the posterior: $P(\lambda^c|\hat{R_0})$
3. New random draw for λ' from the "jump" distribution, which has a mean of the current parameter value, λ^c
   - The most common jump distribution is the Normal distribution

   $$J(\lambda'|\lambda^c) = N(\lambda'|\lambda^c)$$

4. Evaluate the posterior: $P(\lambda'|\hat{R_0})$
5. Decide whether to accept of reject the new value. For example, the accept criteria, $a = \frac{\frac{p(\lambda')}{J(\lambda'|\lambda^c)}}{\frac{p(\lambda^c)}{J(\lambda^c|\lambda')}}$, we will

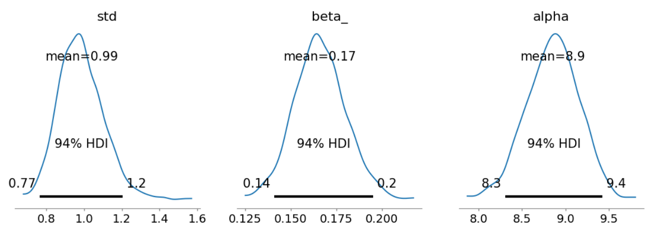
   Accept: $a > 1$

   Reject: $a \leq 1$

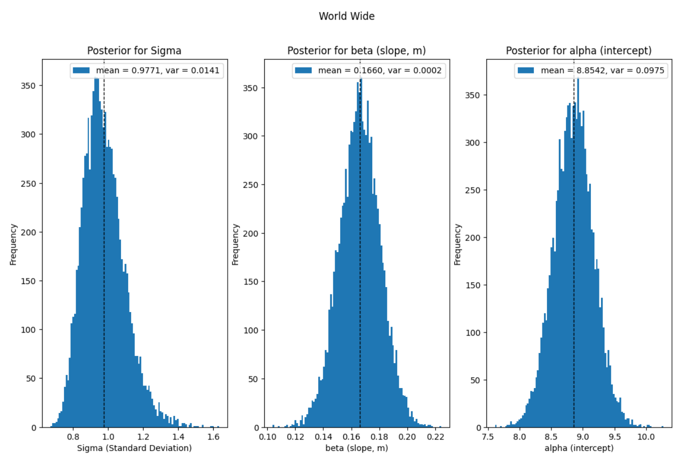
   where to **reject** is to keep the current value of λ^c In practice, this may look something like:

   (a) Initialize: $\lambda^c = \lambda_1$
   (b) For $t = 1, 2, ..., N$
      - $\lambda' = \lambda^c + N(0, 1)$
      - $a = \frac{\frac{p(\lambda')}{J(\lambda'|\lambda^c)}}{\frac{p(\lambda^c)}{J(\lambda^c|\lambda')}}$
      - If $a > 1$
        - **Accept:**$\lambda^c = \lambda'$
      - Else
        - **Reject:** $\lambda^c = \lambda^c$

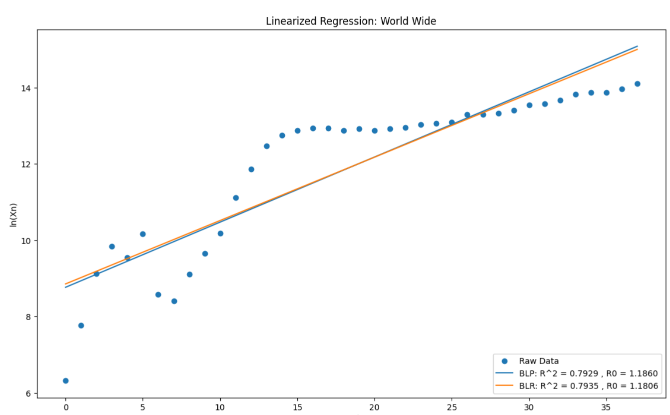
4.2.3 COVID-19 Results Globally & in Los Angeles County, CA.



(a) World Wide: Posterior Distributions of Parameters using MCMC Method

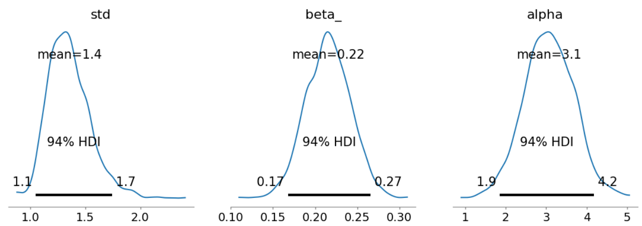


(b) World Wide: Posterior Distributions of Parameters using Objective (Multivariate Normal) Method

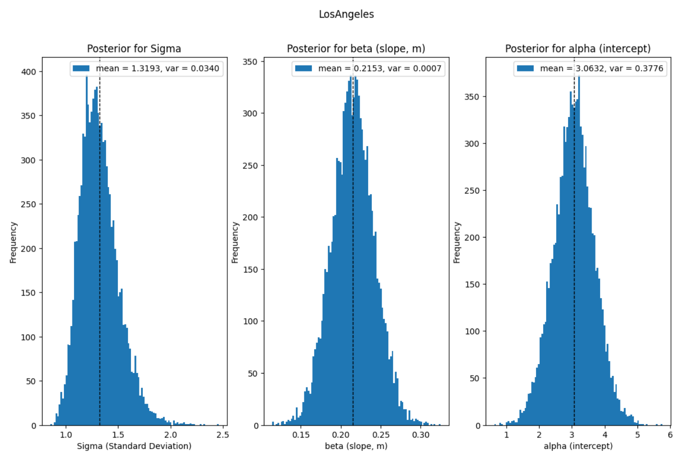


(c) World Wide: Comparison of Best Linear Predictor (BLP) Vs. Objective Bayesian Linear Regression Method (BLR)

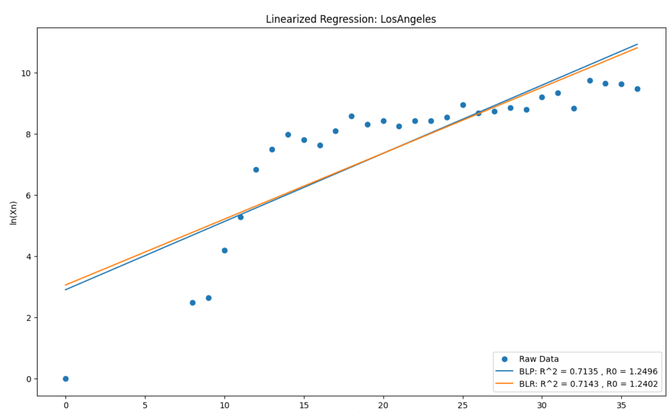
Figure 15: World Wide COVID-19 Bayesian Regression Methods.



(a) Los Angeles: Posterior Distributions of Parameters using MCMC Method



(b) Los Angeles: Posterior Distributions of Parameters using Objective (Multivariate Normal) Method



(c) Los Angeles: Comparison of Best Linear Predictor (BLP) Vs. Objective Bayesian Linear Regression Method (BLR)

Figure 16: Los Angeles County COVID-19 Bayesian Regression Methods.

As we discussed earlier, Berks, Philadelphia, and Manhattan (see Figures 11 & 12) are not particularly linear– they in fact seem to have two distinct sections, where the slope changes drastically. As we'll see in the next section, this is because there is strong evidence of switch points, τ, in these counties. That is, there is evidence that a certain generation, n, the R_e of COVID-19 in these counties changes drastically.
For reasons discussed in the subsequent sections, this does not seem to be the case for Los Angeles county, or COVID-19 globally (see Figures 15 & 16). So, these data sets are good candidates for continuing our exploration of linear regression.

Recall that our linear regression model takes the form

$$\ln(x_n) = n \cdot \ln(\hat{R_0})$$

where x_n represents our observed values of $X(n)$, the number of newly infected each generation and $\hat{R_0}$ our observed values for R_0. Generally regression lines will have the form

$$R = \alpha + \beta \cdot x$$

and $\beta = \ln(\hat{R_0})$ is the slope of the regression line while α is the intercept of the regression line.
In the global case, we can see that the posterior distributions for both methods of Bayesian linear regression (BLR) yield fairly similar results, and for the slope, $\beta = \ln(\hat{R_0})$, and intercept, α, yield means of

$$\beta = \ln(\hat{R_0}) = 0.17\text{, } \alpha = 8.9$$

$$\beta = \ln(\hat{R_0}) = 0.1660\text{, } \alpha = 8.8542$$

for MCMC method and the objective Bayesian regression techniques, respectively.
And, these compare well to the Best Linear Predictor (BLP) results, which yielded:

$$\beta = \ln(\hat{R_0}) = 0.1706$$

$$\alpha = 8.7661$$

And we can see similar the two lines BLP and BLR produce are in Figures 15(c) and 16(c) above, and we can see that both BLP and BLR estimate about an $R_0 = 1.18$, globally for the duration of this data (that is, from 1/22/2020 to 7/27/2020). (Note: for the sake of clarity in figure (c), we excluded the MCMC regression line.)

Similar parity of results can be in the case of Los Angeles County, CA. For the MCMC and objectibe BLR methods, respectively, we see parameters of

$$\beta = \ln(\hat{R_0}) = 0.22\text{, } \alpha = 3.1$$

$$\beta = \ln(\hat{R_0}) = 0.2153\text{, } \alpha = 3.0632$$

and for the BLP method

$$\beta = \ln(\hat{R_0}) = 0.2228$$

$$\alpha = 2.909$$

We can also examine the posterior distributions of the parameters, and see thatthe resultant variance is roughly the same in the MCMC and objective BLR methods, and that these variances are fairly low. We see $Var\sigma = 0.0340$, $Var\beta = 0.0007$, and $Var\alpha = 0.3776$. The variance of β is especially low, which means we can be more certain the true value of β, our slope, and therefore our prediction of R_0

$$\hat{R_0} = e^{\beta}$$

are well fit and well explained by the available data.

It is also interesting to consider what these parameters represent in the real world. We have already discussed this for $\beta = \ln(\hat{R_0})$. The parameter β enables us to find and estimate for R_0, which represents the average number of people an infected individual infects in the next generation, or in the case of our Galton-Watson branching process model: $Y_k^{(j)} \in Po(R_0)$.
The parameter α is similar. We know $\ln(x_n) = \alpha$ when $n = 0$, so $\ln(x_0) = \alpha$. Therefore,

$$x_0 = e^{\alpha}$$

gives us a potential estimate for the size of the initial population of infected in each county or globally.
In both BLP and BLR we also see reasonably high R^2 values of 0.7929 and 0.7935, respectively, in the global

case, and 0.7135 and 0.7143, respectively, in the case of Los Angeles county. According to Duke University [9], R^2 is the percent of the data's variance explained by the regression model, or, in other words, "the fraction by which the variance of the errors is less than the variance of the dependent variable". This means that about 75% and 71% of the variance in our $\ln(x_n)$ data is accounted for by some form of linear regression, and therefore gives some credibility to our predictions of $R_0 = 1.18$ in the global case and $R_0 = 1.20$ in the Los Angeles case– both of which are similar to the currently accepted values for the R_0 of COVID-19 discussed earlier.

5 Preventive Measures During the COVID-19 Pandemic.

5.1 The Difference in λ Before & After Lockdowns.

The book *Bayesian Methods for Hackers* [1] introduces the idea of using Bayesian statistics and and Markov Chain Monte Carlo–via the Python package, PyMC– to determine if the parameter of a random variable being used to model data changes over time. In particular, we are going to be examining the COVID-19 data from the counties of interest – Manhattan, New York, Philadelphia, Pennsylvania, Berks, Pennsylvania, and Los Angeles, California – to see if we can determine if a switch point occurs for R_e. Recall that we earlier demonstrated that

$$R_e = \lambda$$

where $Y_k^{(j)} \in Po(\lambda)$ represents the number of new cases causes by (i.e., in Galton-Watson terms, the offspring of) the k^{th} individual in generation j. This means, we are interested in whether λ changes as the pandemic progresses in these four counties. To do this, we are going to return the linearized data used in the previous section. From this, we can glean spot estimates of R_e for each county, at each generation. This will allow us to have data in the form of $\hat{R}_e$ vs. generation. So, recalling m means the slope, from the previous section we have

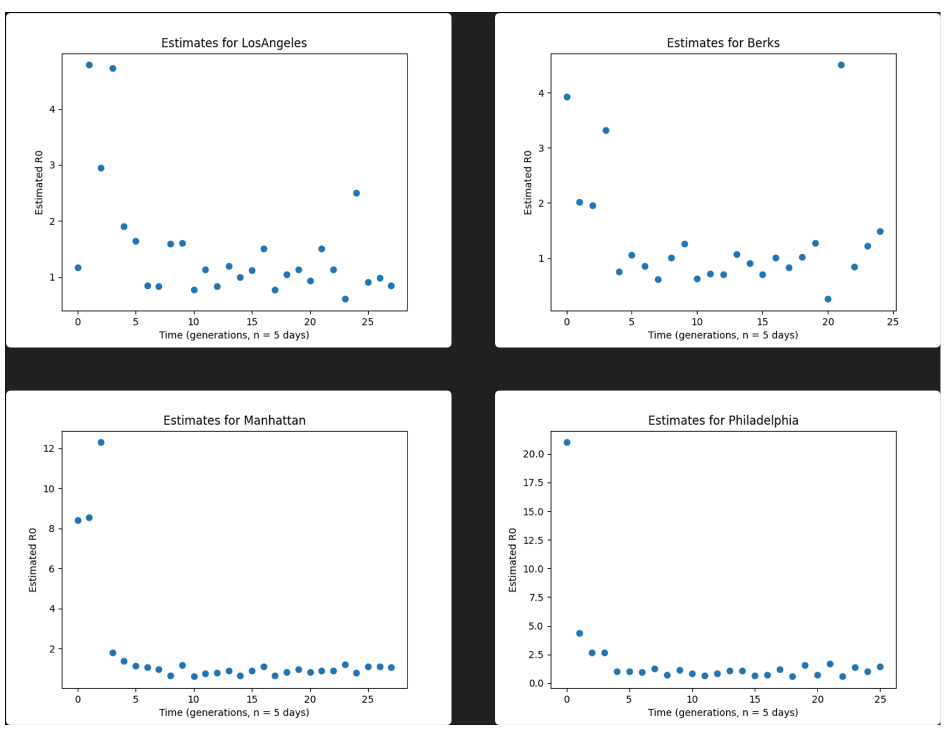
$$m = \ln(\hat{R}_e) \iff \hat{R}_e = e^m$$

This means a spot estimate of $\hat{R}_0$ at each generation can be found simply from

$$(\hat{\lambda}^{(j)} =)\hat{R_e}^{(j)} = \exp\left(\frac{\ln(X(j+1) - \ln(X(j)))}{1 \text{ generation}}\right)$$

where $X(j)$ is the number of newly infected in generation j. We can see below the spot estimates of R_e over time in each county:

Figure 17: $\hat{R}_e$ estimation for for all generations during COVID-19 Pandemic (1/22/20-7/27/20)



As we have already seen through out this project so far, and as is noted by *Bayesian Methods for Hackers*[1], Poisson distributions are appropriate for estimating this kind of count data. So, we will assume R_0

follows a Poisson distribution– in other words, $R_0 \in Po(\epsilon)$, and the number of newly infected per individual generation is Poisson-distributed, with parameter ϵ.

Now, for our switch point, we are assuming

$$R_e = \begin{cases} (\lambda_1 =) R_e^{(1)} & t < \tau \\ (\lambda_2 =) R_e^{(2)} & t \geq \tau \end{cases}$$

Where τ represents the generation at which R_e (noticeably) changes. As for our prior distributions for λ_1 and λ_2, we will use an exponential distribution with parameter $\alpha = \frac{N}{\sum_k \hat{R_0}_k}$. So, as for priors we have [1]

$$(\lambda_1 =) R_e^{(1)} \in Exp(\alpha)$$

$$(\lambda_2 =) R_e^{(2)} \in Exp(\alpha)$$

$$\tau \in U(0, N-1)$$

Then, we will use Markov Chain Monte Carlo (MCMC) to determine the likelihood of each generation being the switch point. This is relatively straight froward and looks like:

1. Assign each variable the appropriate prior
$$P(\lambda_1)$$
$$P(\lambda_2)$$
$$P(\tau)$$

2. Assume each generation is the switch point, τ, and in each case, assign each generation and corresponding $\hat{R_0}^{(j)}$ the appropriate value of λ.

3. For each potential switch point, determine the posterior probabilities
$$P(\lambda_1 | \{\hat{R_e}\})$$
$$P(\lambda_2 | \{\hat{R_e}\})$$
$$P(\tau | \{\hat{R_e}\})$$
by using MCMC, specifically the Metropolis-Hastings algorithm, to sample from our model.

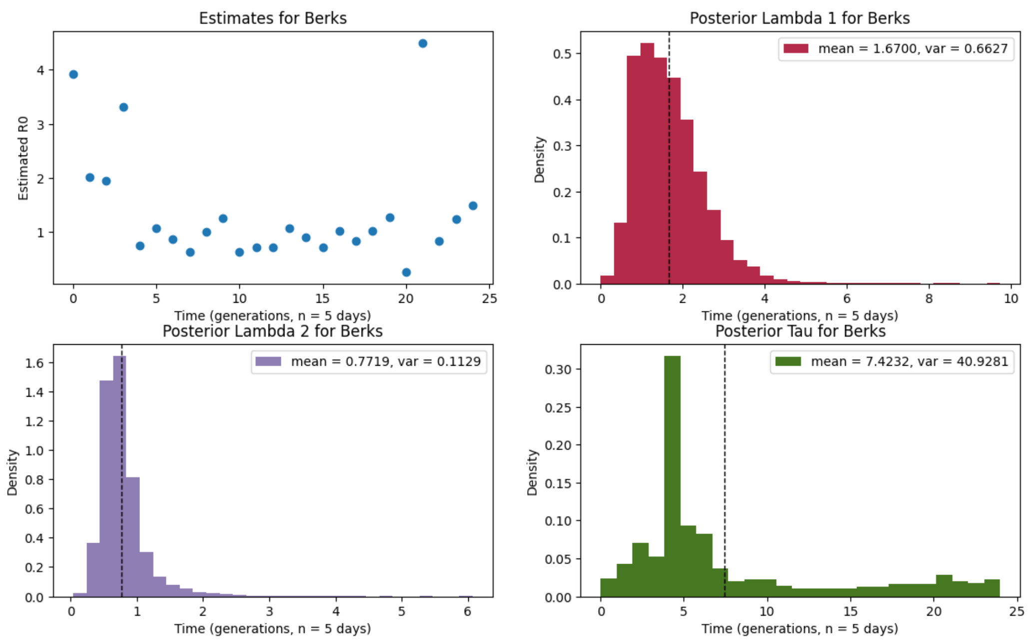
We can then use the generated probabilistic data to plot the posterior distributions of all our unknown parameters– in this case, λ_1, λ_2, and τ– and determine the mean and variance of those unknowns. In the next section, we will briefly discuss MCMC, and lay out the algorithm at a high level, to offer some intuition.

5.1.1 Discussion of COVID-19 Results.

We can see from our spot estimates of $\hat{R_e}$ charts for each county that the initial generation seemingly have extremely high $\hat{R_e}$ values. Intuitively, this makes sense, since COVID-19 cases in these counties were not being tracked until January 22, 2020, but as the name COVID-19 insinuates, cases began springing up in 2019– before the CDC began counting. This would naturally make the initial generation's population, $X(1)$, will seem much higher and therefore the spot estimate of $\hat{R_e}$ at this generation will be artificially inflated. Because of this, we will remove the first $\hat{R_e}$ value from consideration in or A/B testing.

In fact, in some of the cases, the first few generations seem to have abnormally high $\hat{R_e}$ spot estimates. This could be for a variety of reasons– a simple idea is that as COVID-19 was becoming more well known, and influx of people went to get tested in the first several generations (i.e., the first months of 2020) which included newly infected people and people with older infections, whereas as the pandemic progressed, only people with truly new infections were being counted.

Figure 18: A/B Testing Berks, PA



Around the third generation in both the cases of Philadelphia county and Berks county, something happens where the steep incline originally seen in cases suddenly drops off. Both of these counties are in Pennsylvania, and the first reported cases of COVID in Berks and Philadelphia were March 18, 2020 and March 10, 2020, respectively [41]. Given that our generations are 5 days, the fourth generation (20 days after patient 0 is detected in each county) is early April– this is (roughly) where we start to see the slopes of each graph drop off. According to ABC27, the stay-at-home orders in Philadelphia county began **March 23, 2020**, and then Governor Tom Wolff placed all the state on stay-at-home orders on **April 1, 2020** [31]. This begs the questions: **did lock-downs decrease** R_e? Based on the graphs it seems so– in fact, the slopes seem to be **negative** after $n = 4$, and a negative slope on these graphs implies an $R_e < 1 \iff \eta = 1$. We will explore this more in the next section, again using Baye's Analysis to see if λ before and after lock-downs is actually different.

Berks County

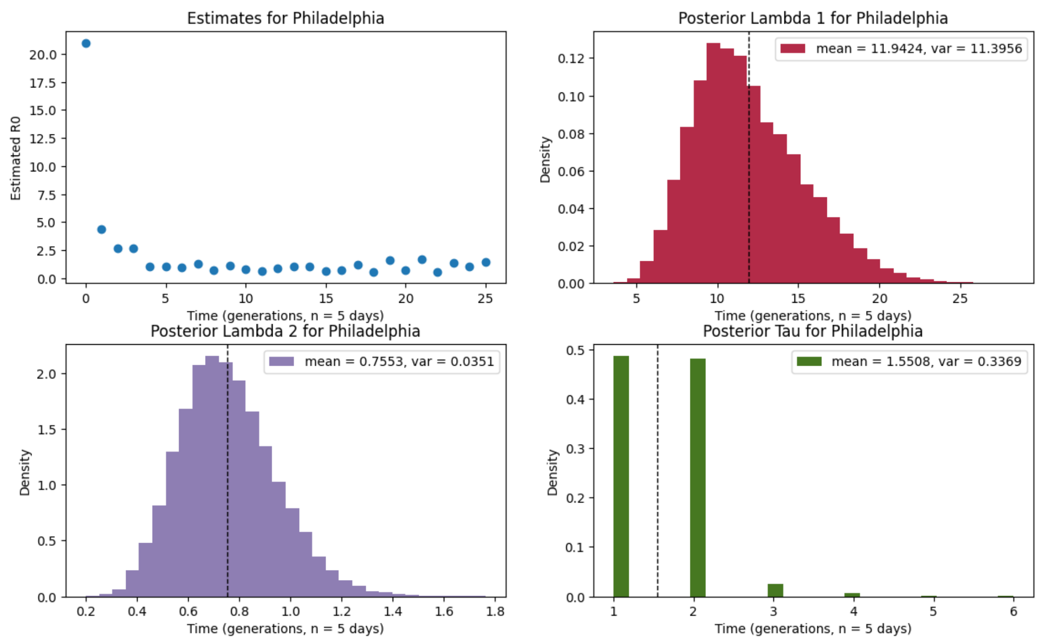
As aforementioned, the first case of COVID-19 reported in Berks County was **March 18, 2020**. Therefore, this is Berks County's Eve, or

$$X(1) = 1$$

at generation $n = 1$. So, when then-Governor Tom Wolff declared a stay-at-home order of **April 1, 2020**, 14 days after Eve, it was around generation $n = 4$. This is a really interesting result, looking at Figure 18, which displays all the A/B testing charts for Berks County, and looking at the posterior distribution of τ, we can see the peak from $n = 4$ to $n = 5$. This tells us that the expected switch point occurred around $n = 4$, corresponding with Pennsylvania's statewide lock-down. We can also see that the posterior distribution of $(R_e^{(1)} =)\lambda_1 = 1.67$ which is around the accepted R_0 value of COVID. But, after the the lockdown, the posterior of $(R_e^{(2)} =) = \lambda_2 = 0.77$ which is significantly lower. In fact, since $(R_e^{(2)} =) = \lambda_2 = 0.77 < 1$, we know from Ahlberg's paper [23], that this is sufficient for extinction–$\eta = 1$– of the disease, in theory.

Philadelphia

Figure 19: A/B Testing Philadelphia, PA



Interestingly, other than the large peaks at the beginning spot estimations of $\hat{R}_e$, the estimates $\hat{R}_e$ do not seem to vary much with time. This is interesting, as the city a devised stay-at-home or lock-down orders fairly early into the outbreak. Taking a look at the CDC data we have been using throughout this project, we can see that the first confirmed case of COVID-19 in Philadelphia County was on **March 10, 2020.** This was Philadelphia's Eve or

$$X(1) = 1$$

At the time, then-Governor Tom Wolff did not lay out a blanket lock-down across the state. Instead, the most populous and most effected counties in south eastern Pennsylvania, to include Philadelphia County, were locked-down on **March 23, 2020**– over a week before Berks County. This means the lock-down occurred around 13 days after the first confirmed case, which is equivalent to around generation $n = 2$ or $n = 3$, bearing in mind our 5 day generations. Turning our attention to Figure 19, which depicts the A/B testing results for Philadelphia County, we can see the peaks at $n = 1$ and $n = 2$. The peak around $n = 1$ is most likely caused by the aforementioned over reporting that is bound to happen in the first several generations. The peak at $n = 2$ is more interesting, and looking at our spot estimates for $\hat{R}_e$, we can see the steady decline in estimates of $\hat{R}_e$ around that time.

Again, as was the case with Berks, we see from the posterior

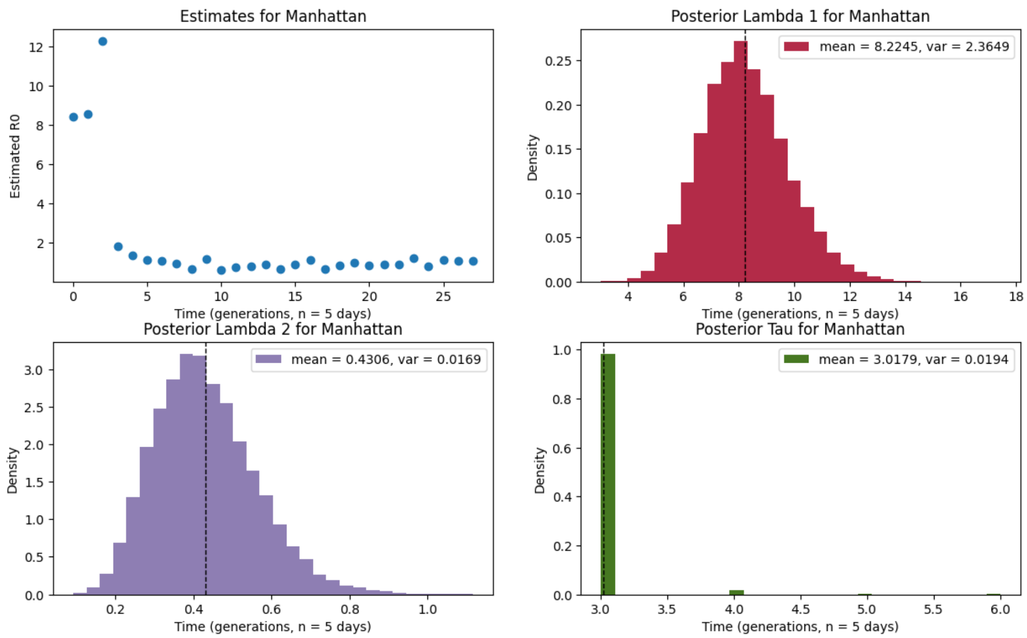
$$(R_e^{(1)} =)\lambda_1 = 11.9424$$

$$(R_e^{(2)} =)\lambda_2 = 0.7553$$

meaning we observe $(R_e^{(2)} =)\lambda_2 = 0.7553 < 1 \iff \eta = 1$ and a possibility of disease extinction. But, because of the high variance of λ_1, these results are slightly less compelling than those of Berks County.

Manhattan. Looking in the CDC data, we see Eve, $X(1) = 1$, occurred on **March 2, 2020.** According to Spectrum News NY1 [33], New York City as a whole did not shut-down everything at once. Instead, the timeline is displayed in the Table below.

Figure 20: A/B Testing Manhattan, NY



Date	Days	Generation n	Event
3/14/20	6	1-2	Public school shutdown
3/17/20	9	2	Bars closed
3/22/20	14	3	Non-essential workers ordered home

This is particularly interesting because the only peak we see on the posterior of τ for Manhattan is $n = 3$ (see Figure 20), which coincides with ordering all non-essential workers to stay home. One could infer a few things from this– it could be argued that:

1. this supports that ordering nearly everybody home is the only way keep

$$(R_e^{(2)} =) = \lambda = 0.43 < 1 \iff \eta = 1$$

   and therefore drive the disease to extinction, or

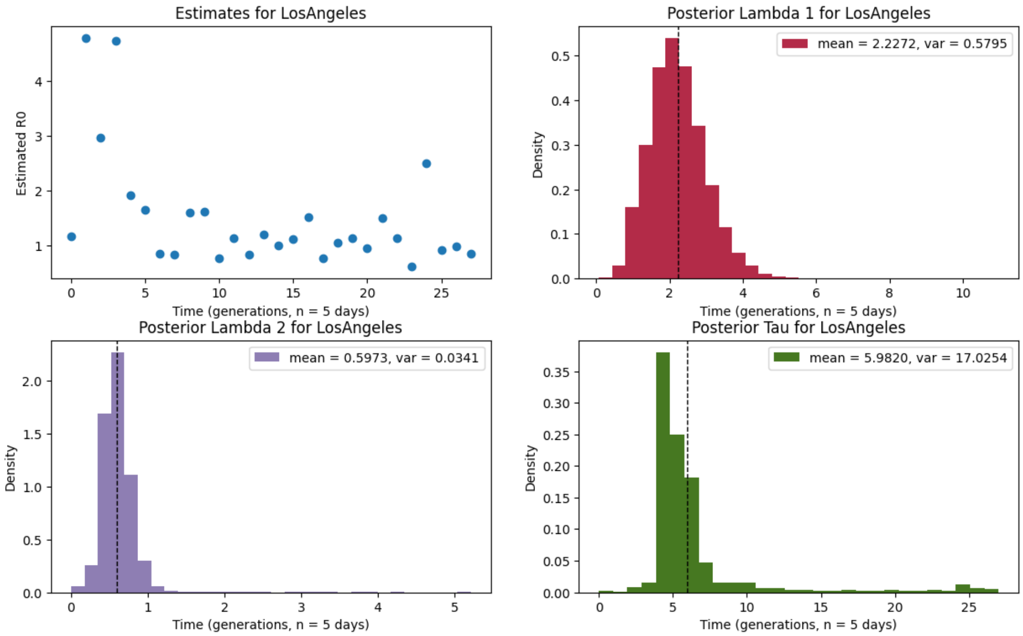
2. that only all 3 of these efforts in tandem enabled a **decrease** in R_e after some switchpoint, τ.

It is also interesting that in this case, the posterior of $(R_e^{(1)} =)\lambda_1 = 8.2245$, which is much higher than the currently accepted COVID-19 R_0 value, but the variance is much lower than what we saw in the case of Philadelphia. This is, again, most likely because of the over reporting in the first few generations of records keeping you would expect to see.

Los Angeles.

The first case of COVID-19 confirmed in Los Angeles County, California, according the CDC data was on **January 26, 2020** (i.e, Eve, $X(1) = 1$). On **March 16, 2020**, 90% of K-12 public school students are ordered to stay home, and then Governor Newsom ordered a stay-at-home order on **March 19, 2020**. Thus, these occurred around $n = 10$ and $n = 10 - 11$, respectively.

Figure 21: A/B Testing Los Angeles, CA



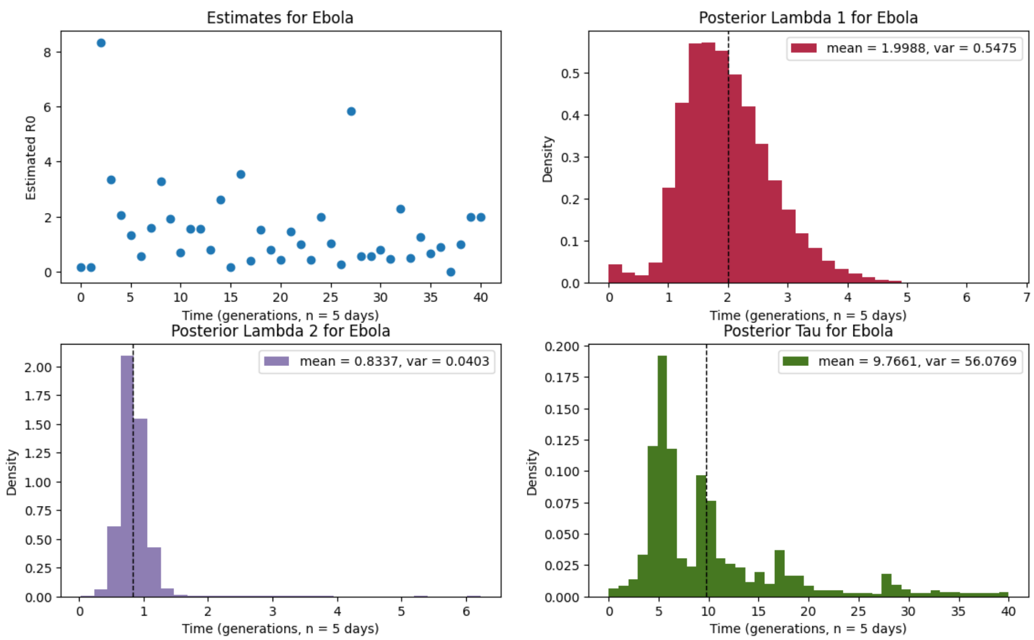
Interestingly, of the four counties we have explored, this is the only one where the peak on the posteriors distribution of τ does not correspond with the being of a shut down (see Figure 21). In fact, the peak is around $n = 5$, or in real time around **February 20, 2020**. This drastically predates the stay-at-home order on March, 19, and it even predates the first action Los Angeles County took against COVID-19– banning social gatherings on **March 7, 2020**[32]– by about 15 days or $n = 3$. It's interesting to note that even in Figure 3, which depicts the the number of new cases by day for each of these four counties, Los Angeles growth is markedly different in shape. Where in Berks, Philadelphia, and Manhattan, we see a steep increase in the number of new cases between generations for the first roughly 10 generations, and then a similar steep descent again into a tail, Los Angeles steadily grows for the whole six month duration of this data.

It's unclear why this is the case, but Los Angeles county did have some exacerbating circumstances the other counties did not. For one, they have four active military bases– Ft. MacArthur, San Clemente Island Naval Test Site, Edwards AFB, and Los Angeles AFB [34]. Military bases are notorious for spreading pandemics and disease. During the Spanish Flu, the initial outbreak can be traced back to a group of First World War soldiers who enlisted in Kansas, than trained in Camp Devens, Massachusetts where one of the first serious outbreaks occurred. Then, sailors from Massachusetts sailed to the Navy Yard in Philadelphia, and the rest is history [36]. The First World War and soldiers in general were responsible for pushing the misnamed Spanish Flu through out the world. Young, predominantly male, populations are mobile and come from all corners of the country, making it difficult to stem the spread, especially since they are considered essential workers.

Summary. In summary, three of the four counties we have examined seemed to have a drastic decrease in $R_e^{(2)}$ corresponding with the implementation of a stay-at-home order, and the one county that did not show this result is an extremely densely populated county with many military service members that could potentially account for the discrepancy. Overall, we have found some evidence for the efficacy of lock-downs as a measure for reducing the spread of COVID-19.

5.1.2 Discussion of Ebola Results.

Figure 22: A/B Testing Ebola Epidemic, West Africa, 2013-2016



The timeline of interventions and actions taken to mitigate the spread of Ebola in West Africa is less clear than the measures taken by individual US states and counties during the COVID-19 pandemic. But, the results are interesting nonetheless. Recall that Ebola has a mean infectious period of approximately 10 days– that is, 1 generation is 10 days. We see by the posterior distributions of λ_1 and λ_2, average R_es of 1.99 and 0.8337, respectively. Again, a $R_e < 1$ corresponds to eventual extinction of an outbreak, $\eta = 1$.

Bearing this in mind, we can see a peak in the posterior distribution of τ around $n = 5$. This is approximately 50 days after the first recorded case in our data. Though this particular epidemic is thought to have begun in late December of 2013 in a village in Guinea, the first date we have data for is **March 25, 2014**, which means our peak in τ is occurring around **May 14, 2014.** There is another peak in the posterior of τ at $n = 100$ or 100 days after March 25, 2014, which was **July 3, 2014.** Interestingly, the second peak on July 3, 2014 corresponds with the CDC deploying o West Africa, where they "assist with response efforts, including surveillance, contact tracing, data management, laboratory testing, and health education" [38].

6 Bayesian Machine Learning for Fatality Prediction.

6.1 Gaussian Naive Bayes Classifier.

In general, Naive Bayes Classifiers are a simple supervised machine learning methods that take advantage of Bayes Theorem to determine the probability of an observation falling into each possible class. They are referred to as "naive", because these models assume **conditional independence** between classification variables– which in reality, is rarely the case. The Gaussian Naive Bayes Classifier makes the additional assumption that each variable is normally distributed [10].

6.1.1 Algorithm & Implementation.

Let $\mathbf{y} = (y_1, y_2, ..., y_n)$ be our observed n responses, and let $\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1k} \\ x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nk} \end{bmatrix}$ be our k observed feature variables, where each each row, $\mathbf{x}_i = (x_{i1}, x_{i2}, ..., x_{ik})$, corresponds to response y_i.

Bayes' Theorem tells us, given class y and x_1 through x_k feature variables

$$P(y|\mathbf{x}) = \frac{P(y)P(\mathbf{x}|y)}{P(\mathbf{x})}$$

And by our assumption of **conditional independence**

$$P(x_i|y, x_1, ..., x_{i-1}, x_{i+1}, ..., x_k) = P(x_i|y)$$

So,

$$P(y|\mathbf{x}) = \frac{P(y)\prod_{j=1}^{k} P(x_j|y)}{P(\mathbf{x})}$$

Naturally, $P(\mathbf{x})$ is constant, and therefore we can treat this equality instead as a proportionality

$$P(y|\mathbf{x}) \propto P(y)\prod_{j=1}^{k} P(x_j|y)$$

From here, in essence the idea is to use this version of Bayes' Theorem to determine the probability of our observed data, $\mathbf{x}$, being of each class in particular, y. This means our final classification prediction boils down to

$$\hat{y} = \text{arg max}_y P(y)\prod_{j=1}^{k} P(x_j|y)$$

where for each class, y, we assume each feature variable, x_i, follows a normal distribution as follows:

$$P(x_j|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(x_j - \mu_y)}{2\sigma_y^2}}$$

[10].

While Naive Bayes' Classifiers are useful in many situations, especially with a small amount of training data compared to other machine learning methods. These type of models are extremely fast compared to other methods, and because of the conditional independence assumption, it allows for probability estimates of each class as 1-dimensional distributions.Interestingly, though Naive Bayes' Classifiers are generally regarded as good classifiers, they are known to be bad estimators, and the resultant probabilities, $P(x_i|y)$ for $1 \leq i \leq k$ are not to be taken as fact [10].

6.1.2 Permutation Importance.

Permutation importance is a simple method for determining how influential each feature variable is in the ultimate classification of data novel to the model.

First, we take a baseline metric, in this case accuracy, on some data set not used to construct the Gaussian Naive Bayes' Classifier model, **X** [11]. Then, we permute each feature column from **X**– that is, the values of the predictor are randomly reassigned to different indices, while the rest of **X** remains unchanged– and the metric is evaluated again. So, permutation importance is the difference between the baseline metric and permutation metric [12]. Thus, we are breaking the relationship between that feature and the corresponding response to see if it matters when calculating our metrics [12]. In the case of Gaussian Naive Bayes Classifiers, the default scoring metric is simply accuracy, and this is the metric we will evaluate.

This is method will use to determine predictor importance in our Gaussian Naive Bayes' Classifier Model.

6.1.3 COVID-19 Severity Risk Factor Results.

Using the Sci-Kit Learn GaussianNB class available in Python, and the accompanying examples and documentation [10], models of these classifiers are very easy to construct.

In the case of COVID-19, we will be using a case severity data set that tracks the associated socioeconomic and health markers of various cases, as well as the ultimate severity of the case [46] [35].

For our purposes here, we want to build a **binary** classification model. In the COVID-19 severity data set [46], the ultimate severity of each case is measured on a scale from 1 to 10, with 10 being the most severe. To this end, we will split our responses as follows: severe cases will be called such for severities ≥ 6, and the cases will be categorized as not severe otherwise. We want to determine what variables are most associated with severe COVID-19 cases.

As we can see from the table below, our Gaussian Naive Bayes' Classifier has an accuracy of 0.8217. The model splits the data using the typical 70-30 split for training and testing, and therefore this classifier was able to correctly classify 82.17% of the COVID-19 cases in the test set. We can also see the permutation importances of each feature in descending order. The top five most important features– that is, the five that change the model's accuracy the most– are PltsScore (platelet count in $\frac{1000}{mm^3}$), Procalctonin concentration, Creatinine (a metabolic waste product) concentration, Ferritin (a blood component containing iron), and blood O_2 saturation < 94 [35].

In other words, according to our Gaussian Naive Bayes' Classifier, these are the biological markers most indicative of an individual who will experience a severe COVID-19 case. The permutation importance reported next to each variable represents the mean change in each variables accuracy after j permutations, and $+-$ the standard deviation after those j permutations.So, even though platelet count (PltsScore) had the greatest impact on accuracy when shuffled varied by $< 1\%$. Because our accuracy is relatively high, sitting at 82% for our testing data, we can conclude that in fact all the features are relatively indicative of the severity of COVID-19 cases.

COVID-19 Severity Risk Factors Permutation Importance.

```
Accuracy: 0.8217821782178217
PltsScore0.009 +/-    0.002
Procalcitonin0.008 +/-    0.002
Creatinine0.007 +/-    0.002
Ferritin0.006 +/-    0.002
O2 Sat < 940.005 +/-    0.002
BUN     0.005 +/-    0.003
```

```
Death   0.004 +/-    0.002
>60     0.003 +/-    0.002
Ddimer  0.003 +/-    0.003
MAP < 700.003 +/-    0.003
INR     0.002 +/-    0.001
Latino  0.002 +/-    0.001
White   0.002 +/-    0.001
Troponin > 0.10.002 +/-    0.003
Plts    0.002 +/-    0.001
DM Complicated0.002 +/-    0.001
BUNYes  0.001 +/-    0.001
IL6     0.001 +/-    0.001
PVD     0.001 +/-    0.001
Lymphocytes < 10.001 +/-    0.001
LOS_Y   0.001 +/-    0.001
LOS     0.001 +/-    0.001
>80     0.001 +/-    0.002
Temp    0.001 +/-    0.002
Asian   0.001 +/-    0.001
CVD     0.001 +/-    0.001
Sodium < 139 or > 1540.001 +/-    0.002
MAP     0.001 +/-    0.002
Stroke  0.001 +/-    0.001
DM Simple0.001 +/-    0.001
CHF     0.001 +/-    0.001
Lympho  0.001 +/-    0.001
ALT     0.001 +/-    0.002
CrctProtein0.001 +/-    0.003
Glucose <60 or > 5000.000 +/-    0.001
WBC     0.000 +/-    0.002
All CNS 0.000 +/-    0.001
AST > 400.000 +/-    0.002
MI      0.000 +/-    0.001
ALT > 400.000 +/-    0.001
Temp > 380.000 +/-    0.000
D-Dimer > 30.000 +/-    0.003
INR > 1.20.000 +/-    0.003
Derivation cohort0.000 +/-    0.001
Seizure 0.000 +/-    0.001
Black   0.000 +/-    0.001
PltsYes 0.000 +/-    0.002
DEMENT  0.000 +/-    0.000
Renal Disease0.000 +/-    0.000
AST     -0.000 +/-    0.001
WBCYes  -0.000 +/-    0.003
LymphoYes-0.000 +/-    0.003
COPD    -0.000 +/-    0.001
ProCalCYes-0.000 +/-    0.001
OldSyncope-0.000 +/-    0.001
CrtnYes -0.000 +/-    0.003
OtherBrnLsn-0.000 +/-    0.001
IL6Yes  -0.000 +/-    0.001
C-Reactive Prot > 10-0.000 +/-    0.003
OsSats  -0.001 +/-    0.002
```

```
Procalciton > 0.1-0.001 +/-    0.002
Glucose -0.001 +/-    0.001
Troponin-0.001 +/-    0.001
ASTYes  -0.001 +/-    0.002
GlucoseYese-0.001 +/-    0.001
O2SatsYes-0.001 +/-    0.001
TempYes -0.001 +/-    0.002
BUN > 30-0.001 +/-    0.003
FerritinYes-0.001 +/-    0.002
WBC <1.8 or > 4.8-0.001 +/-    0.001
OldOtherNeuro-0.001 +/-    0.001
MapYes  -0.001 +/-    0.002
IL6 > 150-0.001 +/-    0.002
Ferritin > 300-0.002 +/-    0.002
Sodium  -0.002 +/-    0.002
Pure CNS-0.002 +/-    0.001
ALTYes  -0.002 +/-    0.002
CrctProtYes-0.003 +/-    0.001
SodimuYes-0.003 +/-    0.002
>70     -0.004 +/-    0.002
INRYes  -0.004 +/-    0.002
DDimerYes-0.004 +/-    0.002
0-60    -0.005 +/-    0.004
TropYes -0.005 +/-    0.002
AgeScore-0.005 +/-    0.003
Age.1   -0.006 +/-    0.003
```

6.1.4 heart failure Risk Factor Results.

As another example of the utility of Gaussian Naive Bayes' Classifiers in mortality prediction in disease, we will also examine a Heart Failure Risk Factor data set [47]. Again we will split this data into training and testing sets using the typical 70-30 split. We can see that we achieved a similar accuracy for prediction in this data set as in the COVID-19 Risk Factor data set, of 0.8333– meaning 83.33% of our test set was correctly classified. As for our permutation importance scores, we see our top five most important features by permutation importance are: ATA (the presence of atypical angina), Up (the presence of an Up-trending ST slope on an ECG), Flat (the presence of an Flat ST slope on an ECG), NAP (the presence of non-anginal pain), and cholesterol level [47]. Interestingly, compared to the COVID-19 data set's most important predictors, the top three predictors for heart failure– ATA and Up or Flat ST slopes– all, on average, varying the accuracy $> 1\%$ after j permutations. The presence of ATA pain especially changes the Gaussian Naive Bayes' Classifiers accuracy by 2.2% on it's own. This indicates that though the presence of ATA pain is the most sindicative predictor of heart failure, given our high overall accuracy of 83.33%, all the features in this data set are potentially important to consider.

heart failure Risk Factors Permutation Importance.

```
Accuracy: 0.8333333333333334
ATA     0.022 +/- 0.010
Up      0.015 +/- 0.013
Flat    0.010 +/- 0.013
NAP     0.006 +/- 0.005
Cholesterol0.005 +/- 0.009
FastingBS0.005 +/- 0.008
ExerciseAngina0.004 +/- 0.010
RestingBP-0.002 +/- 0.003
```

```
LVH     -0.003 +/- 0.003
Age     -0.003 +/- 0.007
Down    -0.003 +/- 0.006
Oldpeak -0.003 +/- 0.011
ST      -0.005 +/- 0.004
Normal  -0.005 +/- 0.004
ASY     -0.005 +/- 0.008
TA      -0.005 +/- 0.004
MaxHR   -0.014 +/- 0.008
M       -0.015 +/- 0.010
F       -0.015 +/- 0.010
```

6.1.5 Conclusion.

The beauty of Gaussian Naive Bayes' Classifiers lie in their simplicity. Though they make assumptions of conditional independence that are not necessarily true in reality, such assumptions make these classifiers easy to compute, use, and interpret. For simple, high dimensional models, these are clearly decent predictors of disease.

6.2 Bayesian Logistic Regression.

6.2.1 Methodology.

The idea of logistic regression is simple: use regression to classify data. For our purposes, let the regression equation be

$$p = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k = \beta \cdot \mathbf{X}$$

We want this to act as our likelihood function (or PDF), $p(x)$– but the issue is that this expression is unbounded, where we need $0 \leq p(x) \leq 1$. So, we will use the logistic function to enforce this range [13]

$$p(x) = \frac{1}{1 + e^{\beta \cdot \mathbf{X}}}$$

For our purposes, we are interested in **binary** classification problems, and so the logistic function will work well. In multi-class classification problems, we can use the generalized form of the logistic function, called the softmax function

$$P(y = j|\mathbf{X}) = \frac{e^{p_k}}{\sum_{k=1}^{K} e^{p_k}}$$

where p is defined as above. The beauty of this that the sum of the likelihoods of each class is 1 [13][14].

The Bayesian aspect of Bayesian Logistic Regression is very similar to the aforementioned Bayesian Linear Regression. For each class, we will determine posterior distributions for each β parameter. We will select an **uninformative**, normal prior with mean 0 and $\sigma \geq 10^4$. Then, we will use MCMC to draw samples from our posteriors, given the disease data sets.

In order to make predictions using the resultant posteriors of β, we will simply use the mean, and report the variances and $\hat{R}$– the measure of convergence for the MCMC of each parameter ($\hat{R}$ approaching 1 indicates convergence).

For both the COVID-19 Risk Factors and Heart Failure Risk Factors data sets, we will use the typical 70-30 split for our training and testing sets and accuracy will be our final fit metric.

6.2.2 Heart Failure Risk Factor Results.

Figure 23: Heart Failure Bayesian Logistic Regression Parameter β, a Posterior Distribution Summaries.

Accuracy: 0.8608695652173913

Summary of Posterior Distributions for all a & β:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
β[0, 0]	4.890	6.165	-2.323	17.940	2.926	2.222	5.0	27.0	1.39
β[0, 1]	8.280	14.513	-5.452	33.110	9.752	8.087	3.0	39.0	2.23
β[1, 0]	-1.933	18.876	-28.780	19.993	12.945	10.825	2.0	39.0	2.95
β[1, 1]	2.857	10.433	-17.809	13.376	6.010	4.746	3.0	15.0	1.92
β[2, 0]	22.834	8.189	6.307	34.611	3.710	2.946	6.0	11.0	1.43
β[2, 1]	24.980	4.875	16.573	35.777	1.615	1.180	9.0	12.0	1.40
β[3, 0]	2.283	8.292	-11.857	21.860	1.888	1.835	14.0	23.0	1.68
β[3, 1]	2.855	8.398	-11.367	22.067	1.837	2.439	23.0	20.0	1.79
β[4, 0]	-5.364	4.473	-13.623	4.472	1.321	0.959	9.0	11.0	1.60
β[4, 1]	-4.867	5.067	-11.976	3.096	3.372	2.893	3.0	22.0	2.28
β[5, 0]	6.558	20.319	-13.790	39.153	13.308	10.930	3.0	12.0	2.05
β[5, 1]	7.377	23.853	-16.747	46.268	15.921	13.169	3.0	22.0	1.92
β[6, 0]	-15.501	20.803	-52.431	10.257	13.241	11.238	3.0	13.0	2.31
β[6, 1]	-5.381	8.092	-24.248	4.854	4.546	3.568	4.0	15.0	1.48
β[7, 0]	46.902	89.625	-59.239	148.533	62.615	52.779	3.0	22.0	2.05
β[7, 1]	56.852	79.523	-31.135	146.035	55.798	47.123	3.0	17.0	1.92
β[8, 0]	-13.293	16.828	-42.232	4.032	10.504	8.487	3.0	14.0	1.94
β[8, 1]	-3.624	9.915	-26.709	8.169	4.129	3.082	6.0	22.0	1.28
β[9, 0]	-0.502	2.217	-3.797	3.250	1.443	1.183	3.0	18.0	2.12
β[9, 1]	-3.360	1.862	-7.336	-0.881	1.174	0.952	3.0	12.0	1.95
β[10, 0]	-0.571	1.338	-3.251	1.494	0.372	0.269	14.0	39.0	1.08
β[10, 1]	-1.867	1.491	-4.246	1.167	0.739	0.565	4.0	15.0	1.42
β[11, 0]	5.057	10.277	-15.132	16.349	6.627	5.412	3.0	24.0	2.31
β[11, 1]	1.123	8.985	-16.625	11.755	5.695	4.625	3.0	23.0	2.26
β[12, 0]	0.858	2.217	-3.386	5.799	1.038	0.820	5.0	22.0	1.77
β[12, 1]	1.960	2.233	-2.065	6.994	1.062	0.933	5.0	22.0	1.93
β[13, 0]	-23.682	22.787	-53.223	3.062	15.946	13.451	2.0	12.0	2.89
β[13, 1]	-23.666	22.787	-53.199	3.085	15.947	13.452	2.0	12.0	2.89
β[14, 0]	0.051	0.107	-0.185	0.219	0.050	0.038	4.0	13.0	1.39
β[14, 1]	0.053	0.108	-0.188	0.221	0.050	0.038	4.0	13.0	1.40
β[15, 0]	5.261	4.969	0.175	10.976	3.489	2.947	3.0	33.0	1.97
β[15, 1]	5.256	4.969	0.169	10.970	3.489	2.947	3.0	33.0	1.97
β[16, 0]	0.523	3.185	-4.289	4.414	2.204	1.851	3.0	15.0	2.09
β[16, 1]	1.546	3.106	-2.799	5.928	2.148	1.802	3.0	13.0	2.13
β[17, 0]	1.409	0.234	1.041	1.810	0.107	0.081	5.0	20.0	1.42
β[17, 1]	1.406	0.234	1.041	1.807	0.106	0.080	5.0	22.0	1.47
β[18, 0]	-7.689	9.561	-21.154	3.718	6.011	4.868	3.0	33.0	1.61
β[18, 1]	-7.234	9.559	-20.742	4.156	6.008	4.865	3.0	35.0	1.61
a[0]	20.234	19.556	-3.384	46.685	13.053	10.797	2.0	11.0	2.34
a[1]	6.330	13.074	-19.631	28.180	8.216	6.653	3.0	15.0	1.93

For reference during this discussion, $\beta[i,j]$ refers to parameter β_i for class j, and $a[j]$ refers to the intercept of p for class j.

The first thing to note is that MCMC has trouble converging for this heart failure risk factor data [47]. We can see this in the relatively high $\hat{R}$ values (ee Figure 23). Typically, $\hat{R} < 1.01$ is considered converged [15], but we are seeing $\hat{R}$s as high as 2.89 for some parameters. This implies we are not confident about the shape of our posterior distributions and are therefore unsure about the mean.

For the sake of time and memory due to the shear number of β parameters needed in this model (19 features x 2 classes + 2 intercepts = 40 β parameters) the number of samples taken by MCMC was only 1000. This could contribute to the overall lack of convergence we are seeings.

All that being said, we are still seeing decent accuracy, with about 86.09% of our test set being correctly classified. This seems to point to the idea that we simply need more samples from the distribution of each β and a parameter in order to see better convergence.

6.2.3 COVID-19 Severity Risk Factors Results.

Figure 24: COVID-19 Severity Bayesian Logistic Regression Parameter β, a Posterior Distribution Summaries.

```
Accuracy: 0.978076379066478
Summary of Posterior Distributions for all a & β:
               mean      sd   hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  ess_tail  r_hat
β[0, 0]       0.020   1.113   -2.375    1.309      0.720    0.589       3.0      14.0   1.67
β[0, 1]      -0.228   1.035   -2.363    0.903      0.684    0.564       3.0      13.0   1.72
β[1, 0]       8.396   7.517   -1.173   20.507      4.671    3.769       3.0      13.0   2.09
β[1, 1]       6.397   7.404   -2.495   18.794      4.602    3.713       3.0      12.0   2.09
β[2, 0]      -0.069   0.159   -0.390    0.260      0.088    0.068       3.0      20.0   1.78
...             ...     ...      ...      ...        ...      ...       ...       ...    ...
β[85, 1]    -29.287  35.391  -89.015    7.811     24.137   20.137       3.0      15.0   2.09
β[86, 0]      7.968  26.993  -27.180   52.417     16.885   13.652       3.0      14.0   1.97
β[86, 1]     -1.787   3.445   -9.645    2.302      1.561    1.295       8.0      16.0   1.78
a[0]        -90.395  98.807 -231.217    1.855     65.667   54.225       3.0      24.0   2.11
a[1]       -214.988  76.702 -386.984  -83.230     41.895   35.352       3.0      11.0   1.65

[176 rows x 9 columns]
```

Because there are 87 features (including one-hot encoded categorical variables), and therefore 176 parameter posteriors, for the COVID-19 severity risk factors data set [46], we cannot display all the parameter posteriors summaries. So, we have an abridged selection in Figure 24.

Like the heart failure risk factor Bayesian logistic regression model, we see very high $\hat{R}$ values, meaning our MCMC sampling did not converge strongly. Also like the heart failure model, we see a very high accuracy. Using the typical 70-30 training-test split, we saw 97.80% correct classification of the test set. Again, this is most likely because of the number of features and the lack of computer power used to build these models, the number of MCMC samples used to build this model was only 1000. With more samples, we could potentially see better convergence, and maintain the high accuracy.

Unlike the heart failure data, the features in the COVID-19 severity data are robust enough that severe cases are very well predicted. An accuracy of 97.80% means a combined instance of Type I and Type II errors (false positives and false negatives, respectively) of only 2.2%.

7 Future Work: Bayesian Belief Network & The Back-Door Path Criterion.

Bayesian Belief Networks (BNNs) are a specific type of **directed acyclical graphs** (DAGs), where the nodes represent variables (in our case, discrete) and the edges represent probabilistic relationships. They are useful for looking at the outcomes of events and determining the likelihood a possible known variables is a contributing factor [**?**].

Using the heart failure risk factor data set [47], we will construct two DAGs and analyze the relationship between our chosen predictors and the occurrence of heart disease. Causal networks are a special case of BNNs [**?**], and so we will be looking at some known causal relationships in our BNNs.

7.0.1 The Back-Door Path Criterion & Calculating Probabilities.

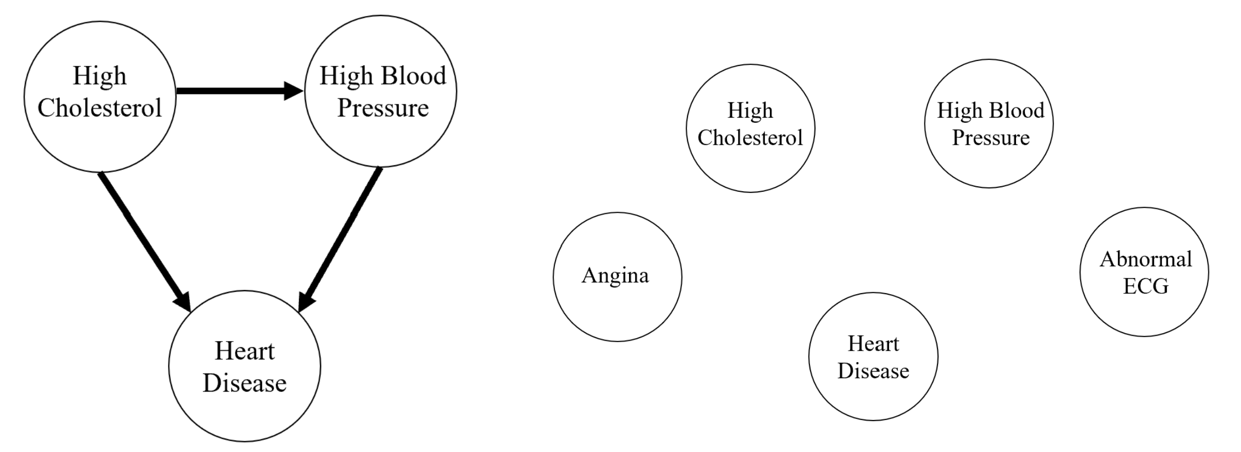
With causal DAGs, the back-door path criterion (BDPC) can be applied. The back-door path criterion is a test that can be applied directly to a causal DAG that tests to see is a set of variables $Z \subseteq V$, where V represents all observed (confounding) variables, is sufficient for identifying $P(y|\hat{x})$, where y is some response and $\hat{x}$ is some specified variable [49]. It states that, relative to a causally ordered pair of variables, (X, Y),

1. no node in Z is a descendant of X
2. Z blocks every path between X and Y that contains an arrow into X

If the back-door path criterion is satisfied, then the causal effect of X on Y is simply

$$P(y|\hat{x}) = \sum_{z} P(y|x,z) \cdot P(z)$$

[49]. In essence, the back-door path criterion allows us to account for the effect of confounding variables.



(a) DAG 1 (b) DAG 2

Figure 25: Directed Acyclical Graphs of Heart Failure Risk Factor Data Set.

Take DAG 1 in Figure 25, as an example. We can see that we have two paths to the response variable "Heart Disease":

High Cholesterol $\rightarrow$ Heart Disease

$$\text{High Cholesterol} \rightarrow \text{High Blood Pressure} \rightarrow \text{Heart Disease}$$

[56].

Let us be interested in the relationship between "High Cholesterol" and "Heart Disease". We need to find $Z \subseteq \{\text{High Cholesterol, High Blood Pressure, Heart Disease}\}$ such that (1) "High Cholesterol" is not a descendant of any node in Z, and (2) Z blocks all paths to "Heart Disease".
(1) is simple, as regardless of what we pick from in {High Cholesterol, High Blood Pressure, Heart Disease} "High Cholesterol" is not a descendant. But, in order to satisfy (2) we need to account for

$$\text{High Cholesterol} \rightarrow \text{High Blood Pressure} \rightarrow \text{Heart Disease}$$

Therefore,

$$Z = \{\text{High Blood Pressure}\}$$

and

$$P(\text{Heart Disease}|\text{High Cholesterol}) = \sum_z P(\text{Heart Disease}|\text{High Cholesterol}, z) \cdot P(z)$$

$$= P(\text{Heart Disease}|\text{High Cholesterol, High Blood Pressure}) \cdot P(\text{High Blood Pressure})$$

Even if the back-door path criterion is not satisfied, we can still use the chain rule of probability to find the joint PDF [**?**] [55]

$$P(\text{Heart Disease, High Blood Pressure, High Cholesterol}) =$$

$$P(\text{Heart Disease}|\text{High Blood Pressure, High Cholesterol})P(\text{High Blood Pressure}|\text{High Cholesterol})P(\text{High Cholesterol})$$

From here on, let

$$C = \text{High Cholesterol}$$

$$B = \text{High Blood Pressure}$$

$$D = \text{Heart Disease}$$

Now, the trick is to identify those conditional probabilities. For that, we will be using the PGMpy library in Python. The goal is to find the **conditional probability distributions** (CPDs) between all the variables involved in DAG 1. There are several ways to accomplish this, one being the **Maximum Likelihood Estimator** (MLE) method [52]. In this case, the MLE would simply be all the relative frequencies in the given data, but, naturally, this method is strongly susceptible to **overfitting** the given data. Another method, which we have discussed elsewhere in this project, is Bayesian parameter estimation.

For Bayesian parameter estimation of the CPDs we will choose the Bayesian Dirichlet equivalent uniform distribution as our prior. The estimated values in the CPDs will be more conservative [52].

For the sake of simplicity, we will transform our cholesterol and blood pressure measurement data in to binary categorical variables, where

$$B = \begin{cases} 1 & \text{blood pressure} \geq {}^{130}\!/_{80} \text{ mmHg} \\ 0 & \text{otherwise} \end{cases}$$

$$C = \begin{cases} 1 & \text{cholesterol} \geq 240 \text{ mg/dL} \\ 0 & \text{otherwise} \end{cases}$$

[56].

We have already established (C, D) conditioned on $Z = \{B\}$ satisfies the BDPC, so we know

$$P(D = 1|C = 1) = \sum_{b \in \{0,1\}} P(D = 1|C = 1, B = b)P(B = b)$$

Figure 26: CPDs for DAG 1.

High Cholesterol(0)	0.347237
High Cholesterol(1)	0.652763

High Cholesterol	High Cholesterol(0)	High Cholesterol(1)
High Blood Pressure(0)	0.45631825273010923	0.38381742738589214
High Blood Pressure(1)	0.5436817472698908	0.6161825726141079

High Blood Pressure	High Blood Pressure(0)	High Blood Pressure(0)	High Blood Pressure(1)	High Blood Pressure(1)
High Cholesterol	High Cholesterol(0)	High Cholesterol(1)	High Cholesterol(0)	High Cholesterol(1)
HeartDisease(0)	0.3188034188034188	0.5864864864864865	0.3651362984218077	0.44882154882154884
HeartDisease(1)	0.6811965811965812	0.4135135135135135	0.6348637015781923	0.5511784511784512

```
High Blood Pressure(0) =  0.4084967320261438
High Blood Pressure(1) =  0.5915032679738562
```

$$= P(D=1|C=1,B=0)P(B=0) + P(D=1|C=1,B=1)P(B=1)$$
$$= (0.4135)(0.4085) + (0.5512)(0.5915)$$
$$P(D=1|C=1) = 0.4950$$

Meaning, that if you walk into the ER with high cholesterol, there is a 49.50% chance you have heart disease.

We can also use the chain rule for probability method, given the CPDs we have from Python. So, given that

$$P(D,B,C) = P(D|B,C)P(B|C)P(C)$$

if we wanted to find the probability of someone walking into a hospital who does not have high blood pressure or high cholesterol, we simply need to compute

$$P(D=1,B=0,C=1) = P(D=1|B=0,C=0)P(B=0|C=0)P(C=0)$$
$$= 0.6812 \cdot 0.4563 \cdot 0.3472$$
$$= 0.1079$$

In other words, if you walk into an ER with normal cholesterol and blood pressure levels, than the probability you have heart disease drops to 10.79%.

We can also work backwards to find things like

$$P(C=1|D=1) = \frac{\sum_{b\in\{0,1\}} P(D=1,B=b,C=1)}{\sum_{b,c\in\{0,1\}} P(D=1,B=b,C=c)}$$

$$= \frac{0.4135 + 0.5512}{0.6812 + 0.4135 + 0.6349 + 0.5512} = 0.4230$$

So, those who come into the hospital with heart disease, have a probability of 42.30% of also having high cholesterol.

DAG 2 (see Figure 25) is a little more complicated. We have added two variables to this DAG: angina and abnormal ECGs. But, there are no causal cycles, and at least in the case of "High Cholesterol" (C), it still satisfies the BDPC. The resultant CPD table is in Figure 27.

Figure 27: CPDs for DAG 2.

High Cholesterol	High Cholesterol(0)	High Cholesterol(1)
Chest Pain(0)	0.5998439937597504	0.508298755186722
Chest Pain(1)	0.4001560062402496	0.491701244813278

Chest Pain	Chest Pain(0)	Chest Pain(1)
Abnormal ECG(0)	0.5722166499498496	0.6342756183745583
Abnormal ECG(1)	0.42778335005015045	0.3657243816254417

High Cholesterol(0)	0.347237
High Cholesterol(1)	0.652763

High Blood Pressure	High Blood Pressure(0)	High Blood Pressure(0)	High Blood Pressure(1)	High Blood Pressure(1)
High Cholesterol	High Cholesterol(0)	High Cholesterol(1)	High Cholesterol(0)	High Cholesterol(1)
HeartDisease(0)	0.3188034188034188	0.5864864864864865	0.3651362984218077	0.44882154882154884
HeartDisease(1)	0.6811965811965812	0.4135135135135135	0.6348637015781923	0.5511784511784512

High Cholesterol	High Cholesterol(0)	High Cholesterol(1)
High Blood Pressure(0)	0.45631825273010923	0.38381742738589214
High Blood Pressure(1)	0.5436817472698908	0.6161825726141079

7.1 Structure Search Methods.

But, in the case of DAG 2, it can be difficult to know whether this DAG is legitimate. That is to say, whether the causal edges we have chosen seem to be true given the data. It could also be the case that we do not know the causal relationship between our variables at all, and that is something we need to glean from the data. Luckily, there are several search algorithms that sift through DAGs in search of the best fit.

The most obvious search method is brute-force. In the case of DAGs, this is called the **Exhaustive Search** method. For the variables in DAG 2, exhaustive search using PGMpy in Python yields

```
Exhaustive Search (BIC): [('Abnormal ECG', 'High Blood Pressure'),

('HeartDisease', 'Abnormal ECG'), ('HeartDisease', 'Chest Pain'),

('HeartDisease', 'High Cholesterol')]
```

as our best fitting DAG. Or,

Of course, exhaustive search becomes unfeasible when there are many features, as the number of DAGs to check grows exponentially. For example, think of the case of the COVID-19 severity risk factor data set, which has 87 features [46]. Instead, we can use **Hill Climb Search**. Hill climb search is a heuristic search, very similar to simulated annealing. Both Hill climb Search and simulated annealing begin with a random solution (in this case DAG) and make small, iterative changes to that solution. Then, they decide whether or not to accept or reject the new solution based on some scoring criteria. The difference between the two algorithms is that simulated annealing is probabilistic, and allows for occasional acceptance of slightly worse solutions in the hopes of exiting a possible **local extrema**, where Hill Climb Search never accepts a worse solution [53]. The scoring criteria used is called **Bayesian Information Criterion** (BIC).

$$BIC = k\ln(n) - 2\ln(\hat{L})$$

where k is the number of features in the model, n is the sample size of the data, and $\hat{L}$ is the maximized value of the likelihood function of the model [50]. Hill Climb search of the variables in DAG 2 using PGMpy in Python yields:

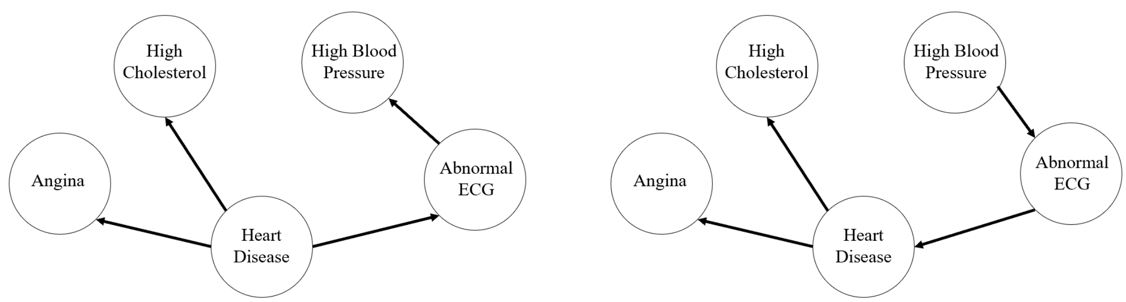
```
Hill Climb Search (BIC): [('Abnormal ECG', 'HeartDisease'),

('High Blood Pressure', 'Abnormal ECG'),

('HeartDisease', 'Chest Pain'), ('HeartDisease', 'High Cholesterol')]
```

The resultant DAGs for each method of search can be seen in Figure 28. The DAG from exhaustive search is on the left, and Hill Climb search on the right.

Figure 28: Proposed DAGs from Exhaustive Search & Hill Climb Search, respectively.



These DAGs are interesting, but some of the causal relationships being posited by these search methods make little sense. For one, in the exhaustive search (left) DAG, an abnormal ECG reading would not cause high blood pressure, and in the Hill Climb search DAG (right) would not cause heart disease– in fact, an ECG would not cause anything. And, also in the Hill Climb search DAG (right), heart disease would not cause high cholesterol– this relationship should be flipped. Given that we are working with an abridges version of the full heart failure risk factor data set (four features out of 11 possible), and a relatively small sample size ($n = 918$), we cannot expect the DAGs found via these search algorithms to be perfect.

In Figure 29, we see a causal DAG that is closer to the true relationships between these variables [56] [57] [58] [59].

When we compare the BIC scores of these three DAGs we see:

i	DAG	BIC Score
A	Exhaustive (Fig. 28, L)	-2977.9408
B	Hill Climb (Fig. 28, R)	-2978.3965
C	Proposed (Fig. 29)	-3108.0928

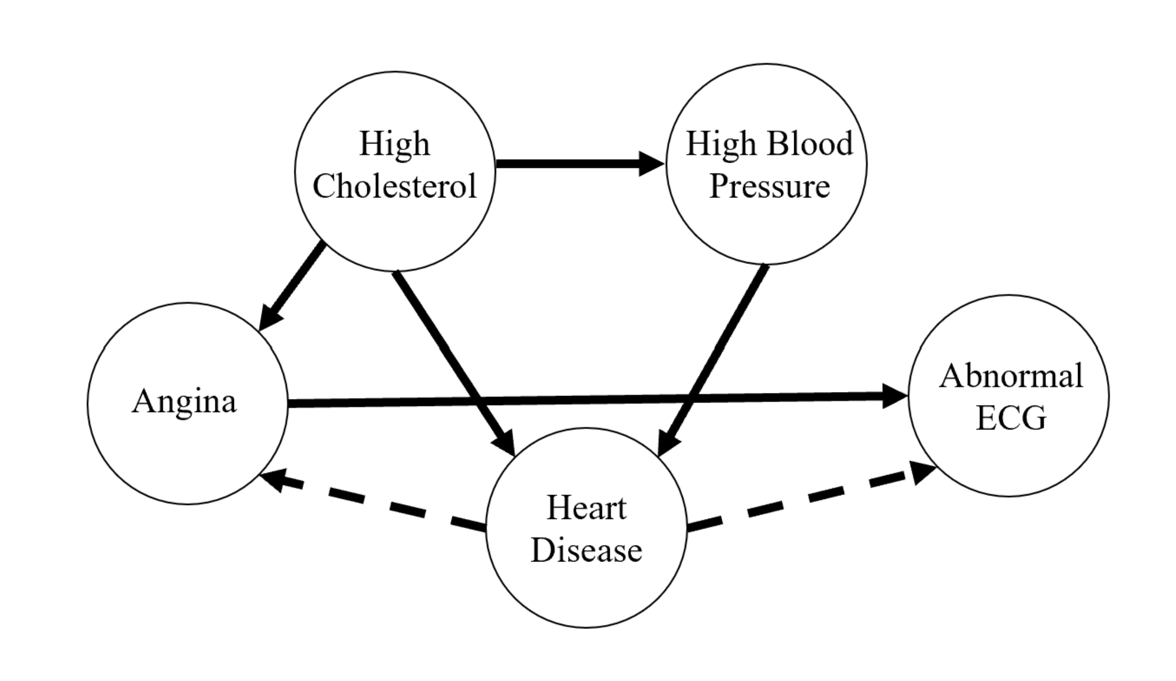
7.1.1 Likelihood Ratio Test.

Just because our DAG search algorithms selected DAGs for us, does not necessarily mean those DAGs fit the data **significantly** better than our proposed model [51].

We can set up the following likelihood ratio test:

$$H_0 : M = M_A$$

Figure 29: Probable DAG 2.



$$H_1 : M = M_C$$
$$\lambda = \frac{\hat{L}_A}{\hat{L}_C}$$

So, our null hypothesis is that our data is better modelled by M_A (or, the DAG selected by the exhaustive search). We reject $H_0 : M = M_A$ if $\lambda < c$, where c corresponds to some α significance [51].

Recall, BIC is given by [50]:

$$BIC = k \ln(n) - 2\ln(\hat{L})$$

Regardless of which of these DAGs we are evaluating, the intercept, $k \ln(n)$, will remain the same. In fact,

$$k \ln(n) = 5 \cdot \ln(918) = 34.11$$

Therefore, the term we care about is difference between $2\ln(\hat{L})$ for each model. We know $\hat{L}$ is the likelihood function

$$\hat{L} = p(x|\hat{\theta}, M)$$

where $\hat{\theta}$ are whatever parameters maximize $\hat{L}$ and M is the model [50] [54].

We can find λ through the difference of our BIC scores for BIC_A and BIC_C

$$BIC_C - BIC_A = k\ln(n) - 2\ln(\hat{L}_C) - (k\ln(n) - 2\ln(\hat{L}_A))$$
$$-3108.0928 + 2977.9408 = 2\ln(\hat{L}_A) - 2\ln(\hat{L}_C)$$
$$-65.076 = \ln(\hat{L}_A) - \ln(\hat{L}_C)$$

$$e^{-65.076} = \frac{\hat{L}_A}{\hat{L}_C}$$

So, we can see that $\lambda = e^{-65.076} \approx 0$. This will be less than most c values, so we can be confident we reject $H_0 : M = M_A$. This means that our proposed DAG 2 in Figure 29 is **not** a significantly worse model than the DAG selected by the exhaustive search algorithm.

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